



Mean-squared exponential stabilization of Takagi-Sugeno fuzzy genetic oscillator networks involving switching control failures: A frame of spatio-temporal discretizations

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ABSTRACT

This article considers global asymptotic and exponential stabilizations in the mean-square sense of space-time discrete Takagi-Sugeno (T-S) fuzzy genetic oscillator networks (GONs) with switching actuator failures and Dirichlet controlled boundaries. It presents some exciting results for the criteria of global asymptotic stabilization in the mean-square sense for the proposed discrete T-S fuzzy GONs via the approaches of the Lyapunov-Krasovskii function, the discrete Wirtinger inequality, and the discrete formula of integration by parts. Besides, our research delves into the exciting possibility of global exponential stabilization in the mean-square sense, which offers a more comprehensive view of stabilized T-S fuzzy GONs than asymptotic stabilization. More critically, this study shows that global mean-square stabilizations of space-time discrete T-S fuzzy GONs can be achieved better by taking the values with the smaller diffusion intensities, smaller coupling strengths and bigger connection weights. Compared to the previous literatures, this paper offers a framework to discuss the issues of global stabilizations for space-time discrete T-S fuzzy GONs with switching actuator failures through the boundary controls, and the results are applicable in a wide range of applications. And finally, an illustrative example is presented to prove the method's validity.

1. Introduction

Genetic oscillator networks (GONs) are complex networks of interactions between genes, proteins, and other molecules that control gene expression and cellular functions. It is a rich tapestry of scientific discoveries and technological innovations spanning over a century. From the foundational principles of genetics and the elucidation of DNA's structure to the advent of high-throughput sequencing and genome editing technologies, each milestone has contributed to our current understanding of how genes are regulated and interact within complex networks. This historical context provides a framework for ongoing research and applications in medicine, biotechnology, and synthetic biology [1–5]. In the field of control of gene expression, GONs regulate the timing, location, and level of gene expression. This ensures that genes are expressed in the right cells at the right times and in the appropriate amounts. In multicellular organisms, GONs regulate the immune response, including the activation and differentiation of immune cells in response

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to pathogens. In addition, GONs are essential for the differentiation of stem cells into specialized cell types during development. They orchestrate the expression of genes necessary for the formation of tissues and organs. GONs are fundamental to the functioning of biological systems, from single cells to complex organisms. Their roles in controlling gene expression, responding to environmental changes, and maintaining cellular processes underscore their importance in both basic biology and applied sciences. The applications of GONs in medicine, biotechnology, agriculture, and environmental science demonstrate their vast potential for addressing some of the most pressing challenges in these fields, please refer to literatures [6–17] for more contents.

Takagi-Sugeno (T-S) fuzzy networks combine fuzzy logic and mathematical modeling to handle complex, nonlinear systems [18, 19]. They are widely used for control, modeling, and decision-making processes. There are the key functions and applications of T-S fuzzy networks. Such as, T-S fuzzy networks are powerful tools for approximating complex nonlinear functions and systems. They can model systems where precise mathematical descriptions are difficult to obtain. T-S fuzzy models facilitate decision-making processes by incorporating human-like reasoning and handling imprecise information, making them suitable for applications where decisions are based on vague or uncertain data. Besides, T-S fuzzy networks are employed in pattern recognition tasks, including classification and clustering, by modeling the relationships between input patterns and output classes. Summarily, T-S fuzzy networks provide a versatile and effective approach to modeling, control, and decision-making in complex and nonlinear systems. Their ability to handle imprecise and uncertain information makes them valuable across a wide range of applications, from industrial control systems and robotics to finance, healthcare, and environmental science. By combining fuzzy logic with mathematical modeling, T-S fuzzy networks offer robust solutions to many real-world problems where traditional methods may fall short [20–26].

Reaction-diffusion networks describe the concentration changes of chemical substances that undergo both chemical reactions and diffusion processes [6,7]. These networks are essential for understanding complex spatiotemporal patterns and dynamics in various scientific fields. Applications of reaction-diffusion networks are widespread, impacting fields such as biology, medicine, chemistry, ecology, material science, physics, engineering, and computational science. By providing a framework to study how local interactions lead to global patterns, these models offer valuable insights and practical solutions to a wide range of scientific and engineering problems. Notably, discrete networks in time and space (DTSNs) [10–12,27–29] are used to study systems where variables change at discrete intervals rather than continuously. These networks are essential in various fields due to their ability to model complex systems with discrete states or events. It possesses many key functions of discrete networks in time and space. Discrete-time models capture the evolution of systems at distinct time intervals, which is useful for systems naturally described by discrete events (e.g., digital systems, population models). In the meantime, discrete-space models represent systems where space is divided into distinct units (e.g., cellular automata, grid-based simulations). Discrete networks often involve systems with a finite set of states, simplifying analysis and computation. They define rules or functions that govern transitions between states over time or space. Finally, discrete models are well-suited for computer simulations, enabling the study of complex systems through numerical experiments. On balance, discrete networks in time and space provide powerful tools for modeling, analyzing, and simulating systems across a wide range of scientific and engineering disciplines. Their ability to handle systems with discrete states and events, coupled with efficient computational algorithms, makes them invaluable for studying complex dynamics and emergent behavior in both natural and artificial systems. In recent times, a number of publications have addressed the subject of GONs characterized by DTSNs, such as, passivity and synchronization [11], almost automorphism [12], etc.

Networks with actuator failures typically refer to systems or networks where components called actuators can fail or switch between different operational states [30–34]. Such networks include that 1) Fault tolerance networks: They are designed to withstand actuator failures by implementing redundancy or backup mechanisms and this ensures that the system can continue functioning even when one or more actuators fail; 2) Adaptive control networks: switching between different actuators allows for adaptive control strategies where the system dynamically adjusts its behavior based on the operational status of the actuators; 3) Fault detection and diagnostics networks: Networks with switching actuator failures often incorporate fault detection and diagnostic algorithms to identify failed actuators and initiate appropriate corrective actions. Networks with switching actuator failures serve critical functions in various industries, providing fault tolerance, adaptive control, resilience, and reliability in complex systems. Their applications span aerospace, automotive, industrial automation, power systems, medical devices, telecommunications, smart buildings, and more, where uninterrupted operation and safety are paramount. These networks highlight the importance of redundancy, adaptive strategies, and fault management in ensuring the continuous functionality of systems despite component failures or disturbances.

The stabilization of networks [35–38] involves ensuring that networked systems remain stable and operate reliably under various conditions. The primary function of network stabilization is to maintain the stability of interconnected systems, preventing instabilities, oscillations, or catastrophic failures that can disrupt operations. Stabilization methods often involve adaptive control strategies that can adjust system parameters or control inputs in real-time to compensate for changes in operating conditions or disturbances. By stabilizing networks, it becomes possible to optimize the performance of interconnected systems, improving efficiency, responsiveness, and overall quality of service. In applications, stabilizing communication networks can maintain connectivity, manage traffic congestion, and prevent network congestion or failures. In transportation networks, stabilizing traffic flow minimizes congestion, delays, and accidents, especially in urban areas and intelligent transportation systems. Applying stabilization methods in railway signaling and control systems ensures safe and efficient train operations. In aerospace and defense, the stabilization of aircraft flight control systems is to ensure stable flight conditions, maneuverability, and safety during takeoff, landing, and flight. For the accuracy, reliability, and mission success, the stabilization methods are usually used in military vehicles, weapons systems, and surveillance platforms. Network stabilization plays a crucial role in ensuring the stability, reliability, and performance of interconnected systems across various domains. Its applications range from power systems and communication networks to industrial automation, transportation, environmental control, aerospace, healthcare, finance, and beyond. Some literatures [13–17] had examined the stabilization of

various GONs, but there is a significant gap in the research on this topic for GONs characterized by DTSNs. We are convinced that the exponential stabilization of T-S fuzzy GONs, as demonstrated by DTSNs, represents a critical research priority in the future.

We are delighted to continue the discourse initiated in our previous deliberations concerning the asymptotic and exponential stabilizations of space-time discrete T-S fuzzy GONs subjected to Dirichlet controlled boundaries and switching actuator failures. Remarkably, the Dirichlet boundaries of the stabilized network are incorporated with the controllers in both the initial and final points of the space variable of the network. The Lyapunov-Krasovskii technique allows us to examine in greater detail the process of network stabilization, from the initial phase of the asymptotic frame to the final stage of the exponential frame [39]. In this section, we will present the most significant findings of this discussion.

- 1) An original frame of discrete T-S fuzzy GONs in the space-time settings, which is involved with switching actuator failures, is set up. The proposed approach represents an advancement over the previous time or space-time discrete GONs [8–12,17], offering a more comprehensive and worthy of attention approach.
- 2) The paper presents an analysis of the topic of asymptotic stabilization for discrete T-S fuzzy GONs through the controllers in the Dirichlet boundaries, which differs from the analysis presented in the published paper [11,27] on Neumann boundaries.
- 3) This represents a significant advance in the field of discrete space and time T-S fuzzy GONs' stabilization according to the T-S fuzzy technique. This approach represents a significant step forward in the field and builds upon existing results for the stabilization of continuous-time GONs [13–16] and discrete-time GONs [17].

This paper is structured as follows. Section 2 is all about the model of space-time discrete stabilized T-S fuzzy GONs involving switching actuator failures, as well as the concepts of asymptotic or exponential quasi-stabilization, and a whole host of handy lemmas. Via the designation of the boundary controls in the GONs, it achieves the global asymptotic and exponential quasi-stabilizations of the equilibrium point of space-time discrete GONs in Section 3. Section 4 provides some illustrative numerical example to substantiate the reliability of the main results. The conclusions of this study are presented in Section 5.

Mathematical symbols and notations: Denote $\mathbb{R} := (-\infty, \infty)$, $\mathbb{R}_0 := [0, \infty)$, $\mathbb{Z}_0 := \{0, 1, 2, \dots\}$, $\mathbb{Z}_+ := \mathbb{Z}_0 \setminus \{0\}$, $\mathbb{Z} = \mathbb{Z}_0 \cup (-\mathbb{Z}_+)$. $\mathbb{R}^N$ denotes N -dimensional real space; I represents some suitable dimensional identical matrix. $\mathbf{E}$ represents the mathematical expectation. Set

$$\Delta_i^2 \xi_n^{[\varpi]} = \Delta_i(\Delta_i \xi_n^{[\varpi]}), \quad \Delta_{c,i}^2 \xi_n^{[\varpi]} = \Delta_i(\Delta_i \xi_n^{[\varpi-1]}),$$

where

$$\Delta_i \xi_n^{[\varpi]} = \frac{\xi_n^{[\varpi+1]} - \xi_n^{[\varpi]}}{l}, \quad \xi_n^{[\varpi]} : \mathbb{Z}^2 \rightarrow \mathbb{R}^n, (\varpi, n) \in \mathbb{Z}^2, l > 0.$$

T-S: Takagi-Sugeno; GONs: genetic oscillator networks; DTSNs: discrete networks in time and space; EETD: exponential Euler time difference; ETD: Euler time difference; MSGAQS: globally asymptotically quasi-stabilized in the mean-square sense; MSGEQS: globally exponentially quasi-stabilized in the mean-square sense.

2. T-S fuzzy GONs in DTSNs' frame

The present section presents a discrete model of the space-time T-S fuzzy GONs with Dirichlet boundary controls and switching actuator failures. Subsequently, the stabilized networks, which are incorporated with the switching actuator failures and the Dirichlet controlled boundaries in both the initial and final points of the space variable, are constructed in accordance with the assumptions regarding the Takagi-Sugeno fuzzy rules with respect to the nonlinear Hill functions. After considering the definitions of asymptotic or exponential stabilization for continuous-time models, we will propose the analogous terms for the T-S fuzzy GONs in DTSNs' frame. In the interim, it presents a series of fundamental lemmas for our consideration as we embark on this endeavor.

2.1. GONs in DTSNs' frame

Let us discuss the following GONs with coupled complex network structure [40] and DTSNs' frame:

$$\begin{cases} \xi_{i,n+1}^{[\varpi]} = e^{-Ah} \xi_{i,n}^{[\varpi]} + A^{-1}(I - e^{-Ah}) \left[\Theta \Delta_{c,i}^2 \xi_{i,n}^{[\varpi]} + B g(\xi_{i,n}^{[\varpi]}) + c \sum_{j=1}^N s_{ij} \Pi \xi_{j,n}^{[\varpi]} + \theta_i \right], \\ \zeta_{i,n+1}^{[\varpi]} = e^{-\tilde{A}h} \zeta_{i,n}^{[\varpi]} + \tilde{A}^{-1}(I - e^{-\tilde{A}h}) \left[\tilde{\Theta} \Delta_{c,i}^2 \zeta_{i,n}^{[\varpi]} + \tilde{B} \xi_{i,n}^{[\varpi]} + \tilde{c} \sum_{j=1}^N \tilde{s}_{ij} \Pi \zeta_{j,n}^{[\varpi]} \right], \end{cases} \quad (2.1)$$

where $(\varpi, n) \in (0, \ell)_{\mathbb{Z}} \times \mathbb{Z}_0$, $\ell > 1$ is an integer, $i = 1, 2, \dots, N$; $\xi_i = (\xi_{i1}, \xi_{i2}, \dots, \xi_{im})^T \in \mathbb{R}^m$ and $\zeta_i = (\zeta_{i1}, \zeta_{i2}, \dots, \zeta_{im})^T \in \mathbb{R}^m$ signify the concentration vectors of mRNA and proteins of the i th node orderly; Θ and $\tilde{\Theta}$ denote the diagonal matrices of diffusion rates orderly; A and $\tilde{A}$ represent degradation rate matrices of the mRNA and protein respectively, which are non-singular and diagonal; $B = [b_{pq}]_{m \times m}$ and b_{pq} is described by

$$b_{pq} = \begin{cases} \tilde{b}_{pq}, & \text{if } q \text{ activates gene } r; \\ 0, & \text{if there is no link between } q \text{ and } r; \\ -\tilde{b}_{pq}, & \text{if } q \text{ represses gene } r, \end{cases}$$

where $\tilde{b}_{pq}$ expresses the basal transcriptional rate; $\tilde{B}$ is a diagonal matrix denoting the translation rate; $g(\zeta_i) = (g_1(\zeta_{i1}), g_2(\zeta_{i2}), \dots, g_n(\zeta_{im})) \in \mathbb{R}^m$ denotes the activation vector-valued function and g_q is described by the Hill function with Hill constant $\kappa_q > 1$, namely, $g_q(u) = \frac{u^{\kappa_q}}{1+u^{\kappa_q}}, \forall u \in \mathbb{R}_0$; $c, \tilde{c}, \theta_i$ are positive constants; $S = [s_{ij}]_{N \times N}$ and $\tilde{S} = [\tilde{s}_{ij}]_{N \times N}$ display the adjacency weights to network (2.1); Π represents the matrix of inner coupling between proteins or mRNAs of the same species; $i, j = 1, 2, \dots, N$. GONs (2.1) possesses the following initial and Dirichlet controlled boundaries:

$$\begin{cases} \xi_{i,0}^{[\varpi]} = \varphi_i^{[\varpi]}, \quad \zeta_{i,0}^{[\varpi]} = \tilde{\varphi}_i^{[\varpi]}, \quad \forall \varpi \in (0, \mathcal{E})_{\mathbb{Z}}; \\ \xi_{i,n}^{[0]} = \mu_{i,n}, \zeta_{i,n}^{[0]} = \nu_{i,n}, \quad \xi_{i,n}^{[L]} = \rho_{i,n}, \zeta_{i,n}^{[0]} = \varrho_{i,n}, \quad \forall n \in \mathbb{Z}_0, \end{cases} \quad (2.2)$$

where $\mu_{i,n}, \nu_{i,n}, \rho_{i,n}, \varrho_{i,n}$ denote the boundary control inputs in the i th node at time n , $i = 1, 2, \dots, N$. In GONs (2.1) and (2.2), it utilizes several efficacious, discrete methodologies, encompassing the exponential Euler time difference and the central finite space difference. Should you wish to learn more, you are invited to consult the relevant papers [27–29].

Remark 2.1. In the context of the control issues of the DTSNs [11,27] and diffusive systems [41,42], it is typical for the Neumann controlled boundaries to be adapted, which are distinct from the Dirichlet controlled boundaries as outlined in (2.2). The current discussion has revealed that Dirichlet controlled boundaries are capable of addressing the issue of stabilizing the GONs in DTSNs' frames.

Remark 2.2. Via the prior work outlined in reports [27–29], we suggest that GONs (2.1) and (2.2) can be viewed as a DTSNs of the GONs with reaction-diffusion terms:

$$\begin{cases} \frac{\partial \xi_i}{\partial t} = \Theta \frac{\partial^2 \xi_i}{\partial x^2} - A \xi_i + B g(\zeta_i) + c \sum_{j=1}^N s_{ij} \Pi \xi_j + \theta_i, \\ \frac{\partial \zeta_i}{\partial t} = \tilde{\Theta} \frac{\partial^2 \zeta_i}{\partial x^2} - \tilde{A} \zeta_i + \tilde{B} \xi_i + \tilde{c} \sum_{j=1}^N \tilde{s}_{ij} \Pi \zeta_j, \\ \xi_i(x, 0) = \varphi_i(x), \quad \zeta_i(x, 0) = \tilde{\varphi}_i(x), \quad \forall x \in [0, \gamma]; \\ \xi_i(0, t) = \xi_i(\gamma, t) = 0 = \zeta_i(0, t) = \zeta_i(\gamma, t), \quad \forall t \in \mathbb{R}_0, \end{cases}$$

where $(x, t) \in (0, \gamma) \times \mathbb{R}_0$ for some $\gamma > 0$; $\xi_i = (\xi_{i1}, \xi_{i2}, \dots, \xi_{im})^T \in \mathbb{R}^m$ and $\zeta_i = (\zeta_{i1}, \zeta_{i2}, \dots, \zeta_{im})^T \in \mathbb{R}^m$, $i = 1, 2, \dots, N$.

Remark 2.3. The exponential Euler time difference (EETD) has been employed in the derivation of GONs (2.1), diverging from the Euler time difference (ETD) utilized in the studies of discrete-time GONs [8,9]. According to the works of papers [43,44], the convergent speed of EETD is superior to that of ETD.

2.2. T-S fuzzy networks

To set up the T-S fuzzy stabilized networks of GONs (2.1) and (2.2), it is necessary to make certain assumptions regarding the equilibrium for GONs (2.1). These assumptions are based on the existence and uniqueness results outlined in reference [12], a reputable academic source on this topic.

(g_1) GONs (2.1) admits a unique equilibrium $(\xi_1^*, \dots, \xi_N^*, \zeta_1^*, \dots, \zeta_N^*)$, where $\xi_i^*, \zeta_i^* \in \mathbb{R}^n$ are constant vectors, $i = 1, 2, \dots, N$.

Under assumption (g_1) and letting $\mathbf{u}_{i,n}^{[\varpi]} = \xi_{i,n}^{[\varpi]} - \xi_i^*$, $\mathbf{v}_{i,n}^{[\varpi]} = \zeta_{i,n}^{[\varpi]} - \zeta_i^*$, then GONs (2.1) can be transformed into

$$\begin{cases} \mathbf{u}_{i,n+1}^{[\varpi]} = e^{-Ah} \mathbf{u}_{i,n}^{[\varpi]} + \frac{I - e^{-Ah}}{A} \left[\Theta \Delta_{c,i}^2 \mathbf{u}_{i,n}^{[\varpi]} + B \tilde{g}(\mathbf{v}_{i,n}^{[\varpi]}) + c \sum_{j=1}^N s_{ij} \Pi \mathbf{u}_{j,n}^{[\varpi]} \right], \\ \mathbf{v}_{i,n+1}^{[\varpi]} = e^{-\tilde{A}h} \mathbf{v}_{i,n}^{[\varpi]} + \frac{I - e^{-\tilde{A}h}}{\tilde{A}} \left[\tilde{\Theta} \Delta_{\tilde{c},i}^2 \mathbf{v}_{i,n}^{[\varpi]} + \tilde{B} \mathbf{u}_{i,n}^{[\varpi]} + \tilde{c} \sum_{j=1}^N \tilde{s}_{ij} \Pi \mathbf{v}_{j,n}^{[\varpi]} \right], \end{cases} \quad (2.3)$$

where $\tilde{g}(\mathbf{v}_{i,n}^{[\varpi]}) = g(\zeta_{i,n}^{[\varpi]}) - g(\zeta_i^*)$, $(\varpi, n) \in (0, \mathcal{E})_{\mathbb{Z}} \times \mathbb{Z}_0$, $i = 1, 2, \dots, N$.

On occasion, the determination of the Hill constant in the Hill function is challenging, which introduces an element of uncertainty into the model of GONs (2.1). To address this issue, the use of the T-S fuzzy techniques [18,19], an effective tool for addressing

uncertainty in models, may prove beneficial. The T-S fuzzy system is designed to convert nonlinear systems into local linear representations, where each output space is approximated by a local linear equation. Thus, the following T-S fuzzy GONs for the error network (2.3) can be derived as shown below.

• **Model Rule r for Node i :**

IF $z_{1,n}^{[\varpi]}$ is $M_{r,1}$, $z_{2,n}^{[\varpi]}$ is $M_{r,2}$, ..., and $z_{o,n}^{[\varpi]}$ is $M_{r,o}$, THEN

$$\begin{cases} \mathbf{u}_{i,n+1}^{[\varpi]} = e^{-Ah} \mathbf{u}_{i,n}^{[\varpi]} + \frac{I - e^{-Ah}}{A} \left[\Theta \Delta_{c,i}^2 \mathbf{u}_{i,n}^{[\varpi]} + B_r \mathbf{v}_{i,n}^{[\varpi]} + c \sum_{j=1}^N s_{ij} \Pi \mathbf{u}_{j,n}^{[\varpi]} \right], \\ \mathbf{v}_{i,n+1}^{[\varpi]} = e^{-\tilde{A}h} \mathbf{v}_{i,n}^{[\varpi]} + \frac{I - e^{-\tilde{A}h}}{\tilde{A}} \left[\tilde{\Theta} \Delta_{c,i}^2 \mathbf{v}_{i,n}^{[\varpi]} + \tilde{B} \mathbf{u}_{i,n}^{[\varpi]} + \tilde{c} \sum_{j=1}^N \tilde{s}_{ij} \Pi \mathbf{v}_{j,n}^{[\varpi]} \right], \end{cases} \quad (2.4)$$

where $(\varpi, n) \in (0, \ell)_{\mathbb{Z}} \times \mathbb{Z}_0$, $M_{r,s}$ represents the fuzzy set and L is the number of model rules; $i = 1, 2, \dots, N$, $r = 1, 2, \dots, L$, $s = 1, 2, \dots, o$; $\mathbf{z} = (z_1, z_2, \dots, z_o)$ shows the premise variable vector.

Remark 2.4. As illustrated in paper [45], if transcription factor q is a repressor of gene r in GONs (2.1), the coupling strength is opposite, meaning that the coupling strength may be negative. In this study, the value of the coupling matrix B will affect the coefficient matrix B_r in T-S fuzzy GONs (2.4).

A locally linear network, which is defuzzified by the T-S fuzzy rules previously described in (2.4), is represented by

$$\begin{cases} \mathbf{u}_{i,n+1}^{[\varpi]} = e^{-Ah} \mathbf{u}_{i,n}^{[\varpi]} + \frac{I - e^{-Ah}}{A} \left[\Theta \Delta_{c,i}^2 \mathbf{u}_{i,n}^{[\varpi]} + \sum_{r=1}^L h_r(z_n^{[\varpi]}) B_r \mathbf{v}_{i,n}^{[\varpi]} + c \sum_{j=1}^N s_{ij} \Pi \mathbf{u}_{j,n}^{[\varpi]} \right], \\ \mathbf{v}_{i,n+1}^{[\varpi]} = e^{-\tilde{A}h} \mathbf{v}_{i,n}^{[\varpi]} + \frac{I - e^{-\tilde{A}h}}{\tilde{A}} \left[\tilde{\Theta} \Delta_{c,i}^2 \mathbf{v}_{i,n}^{[\varpi]} + \tilde{B} \mathbf{u}_{i,n}^{[\varpi]} + \tilde{c} \sum_{j=1}^N \tilde{s}_{ij} \Pi \mathbf{v}_{j,n}^{[\varpi]} \right], \end{cases} \quad (2.5)$$

where $(\varpi, n) \in (0, \ell)_{\mathbb{Z}} \times \mathbb{Z}_0$, $i = 1, 2, \dots, N$;

$$h_r(z_n^{[\varpi]}) = \frac{W_r(z_n^{[\varpi]})}{\sum_{r=1}^L W_r(z_n^{[\varpi]})}, \quad W_r(z_n^{[\varpi]}) = \prod_{s=1}^o M_{r,s}(z_{s,n}^{[\varpi]}) \geq 0,$$

hereby, $M_{r,s}(z_{s,n}^{[\varpi]})$ shows the grade of membership of $z_{s,n}^{[\varpi]}$ for $s = 1, 2, \dots, o$, $r = 1, 2, \dots, L$. It should be noted that $\sum_{r=1}^L h_r(\cdot) = 1$. In accordance with assumption (g_1) and the initial boundary values in (2.2), the initial values and Dirichlet boundaries of T-S fuzzy GONs (2.5) are defined as

$$\begin{cases} \mathbf{u}_{i,0}^{[\varpi]} = \varphi_i^{[\varpi]} - \xi_i^*, & \mathbf{v}_{i,0}^{[\varpi]} = \tilde{\varphi}_i^{[\varpi]} - \zeta_i^*, \quad \forall \varpi \in [0, \ell]_{\mathbb{Z}}; \\ \mathbf{u}_{i,n}^{[0]} = \mu_{i,n} - \xi_i^*, & \mathbf{v}_{i,n}^{[0]} = \nu_{i,n} - \zeta_i^*, \\ \mathbf{u}_{i,n}^{[\ell]} = \rho_{i,n} - \xi_i^*, & \mathbf{v}_{i,n}^{[\ell]} = \varrho_{i,n} - \zeta_i^*, \quad \forall n \in \mathbb{Z}_0, \end{cases} \quad (2.6)$$

where $i = 1, 2, \dots, N$.

Remark 2.5. Upon comparison between the error GONs (2.3) and the T-S fuzzy GONs (2.5), it becomes evident that the sole nonlinear term $B\tilde{g}(\mathbf{v})$, present in the initial portion of the error GONs (2.3), has been linearized in the T-S fuzzy GONs (2.5) to form the term $\sum_{r=1}^L h_r(\cdot) B_r \mathbf{v}$. It is important to note that the factor $h_r(\cdot)$ belongs to the interval $[0, 1]$, which will enable the subsequent calculations to be simpler than those required for nonlinear Hill functions in error GONs (2.3).

Remark 2.6. In papers [11,12], GONs in the space-time discrete frames have been researched. In view of GONs (2.5), this article gives the T-S fuzzy structure of space-time discrete GONs, which are distinct from the GONs in papers [11,12].

Remark 2.7. In the extant literature on the DTSNs [11,27] and diffusive systems [41,42], the controllers are invariably located at the end of the space variable of the models' control boundaries. This paper considers the involvement of both the initial and final points of the space variable of T-S fuzzy GONs (2.5) with the controllers that will be subsequently developed.

Remark 2.8. Recently, the authors [46–48] discussed the fuzzy GONs. Such as, without spatial diffusions, papers [46,47] studied the synchronization control and global dissipativity of GONs with fuzzy operations. In article [48], Qin et al. investigated anti-disturbance

synchronization of GONs governed by reaction-diffusion equations via the Takagi-Sugeno fuzzy model. Clearly, this paper considered the space-time discrete version of the reaction-diffusion GONs, which has significant values on the research fields of numerical computations and engineering applications.

2.3. Actuator failures

A stochastic variable, designated by α_n , is employed to display the dynamics of switching occurring between the control apparatus and the actuator. The value of α_n is constrained to the set $\{1, 2, \dots, J\}$, where J is a positive integer. And

$$\Pr\{\alpha_n = j\} = \pi_j, \quad \forall j \in \{1, 2, \dots, J\}; \quad \sum_{j=1}^J \pi_j = 1, \quad \forall n \in \mathbb{Z}_0.$$

Define

$$X_j(n) = \begin{cases} 1, & \text{if } \alpha_n = j; \\ 0, & \text{if } \alpha_n \neq j. \end{cases}$$

Clearly, $\mathbf{E}(X_j(n)) = \pi_j$ for all $j \in \{1, 2, \dots, J\}$ and $n \in \mathbb{Z}_0$. Besides,

$$j_1 \neq j_2 \Rightarrow \Pr\{\alpha_n = j_1, \alpha_n = j_2\} = 0,$$

and

$$\sum_{j=1}^J X_j(n) = 1, \quad \mathbf{E}(X_{j_1}(n)X_{j_2}(n)) = \begin{cases} \pi_{j_1}, & \text{if } j_1 = j_2; \\ 0, & \text{if } j_1 \neq j_2, \end{cases}$$

where $j_1, j_2 \in \{1, 2, \dots, J\}$.

Failures in actuators can occur due to a variety of reasons, typically categorized into mechanical, electrical, thermal, environmental, and operational factors. Understanding and addressing these potential failure modes can significantly enhance the reliability and lifespan of actuators. To model the failures happening in the actuators, the parameters $\theta_{i,\alpha_n}, \vartheta_{i,\alpha_n} \in (0, 1]$, which denote the implementation rates of the i th actuators with respect to mRNA and proteins, respectively, and are dependent on the switching signals $\alpha_n \in \{1, 2, \dots, J\}$, are designed in the controllers as shown below

$$\rho_{i,n}^{\alpha_n} = \theta_{i,\alpha_n} \rho_{i,n}^{\alpha_n}, \quad \varrho_{i,n}^{\alpha_n} = \vartheta_{i,\alpha_n} \varrho_{i,n}^{\alpha_n}, \quad \forall n \in \mathbb{Z}_0, \quad (2.7)$$

where $\rho_{i,n}^{\alpha_n}$ and $\varrho_{i,n}^{\alpha_n}$, whose values will be determined later, denote the controlled inputs, and they will be arranged to T-S fuzzy GONs (2.5), $i = 1, 2, \dots, N$.

Associated with (2.7), the boundary controls, which ensure the zero boundaries on the starting point 0 of the space variable, are designed by

$$\begin{cases} \mu_{i,n} = \xi_i^*, & v_{i,n} = \zeta_i^*, \\ \rho_{i,n} = \theta_{i,\alpha_n} \rho_{i,n}^{\alpha_n}, & \varrho_{i,n} = \vartheta_{i,\alpha_n} \varrho_{i,n}^{\alpha_n}, \end{cases} \quad (2.8)$$

which induces from (2.6) that

$$\begin{cases} \mathbf{u}_{i,0}^{[\varpi]} = \varphi_i^{[\varpi]} - \xi_i^*, & \mathbf{v}_{i,0}^{[\varpi]} = \tilde{\varphi}_i^{[\varpi]} - \zeta_i^*, & \forall \varpi \in [0, \ell]_{\mathbb{Z}}; \\ \mathbf{u}_{i,n}^{[0]} = 0, & \mathbf{v}_{i,n}^{[0]} = 0, & \mathbf{u}_{i,n}^{[\ell]} = \theta_{i,\alpha_n} \rho_{i,n}^{\alpha_n} - \xi_i^*, & \mathbf{v}_{i,n}^{[\ell]} = \vartheta_{i,\alpha_n} \varrho_{i,n}^{\alpha_n} - \zeta_i^*, \end{cases} \quad (2.9)$$

where $n \in \mathbb{Z}_0$, $i = 1, 2, \dots, N$.

For the error levels of the actuators, the assumption below is important for the later study.

$$(g_2) \quad \theta_{i,j}, \vartheta_{i,j} \in (0, 1], \quad \forall j \in \{1, 2, \dots, J\}, \quad i = 1, 2, \dots, N.$$

Remark 2.9. It should be pointed out that θ, ϑ display how well the i th actuators control mRNA and proteins, respectively. If $\theta \rightarrow 0$, it implies that the actuator tends to the state that is completely invalid. If $\theta \neq 0$, the actuator yields the partial error, which generates a part of control. If $\theta = 1$, the actuator is normal without any errors. For parameter ϑ , it can be understood as the same as θ .

Similar to the definition of stability for (random) difference equations [49,50], the definition of stability for random space-time difference equations is given as described below.

Definition 2.1. Consider the following random space-time difference equations:

$$\begin{cases} \mathbf{x}_{n+1}^{[\varpi]} = f(\mathbf{x}_n^{[\varpi]}), & \forall (\varpi, n) \in (0, \ell]_{\mathbb{Z}} \times \mathbb{Z}_0; \\ \mathbf{x}_0^{[\varpi]} = \psi^{[\varpi]}, & \forall \varpi \in (0, \ell]_{\mathbb{Z}}; \\ \mathbf{x}_n^{[0]} = \tilde{\psi}_n, \mathbf{x}_n^{[\ell]} = \tilde{\psi}_n, & \forall n \in \mathbb{Z}_0, \end{cases} \quad (2.10)$$

where $\mathbf{x}_n$ denotes N -dimensional random variable, $f : \mathbb{R}^N \rightarrow \mathbb{R}^N$ and $f(0) = 0$. Then Eq. (2.10) is said to be globally asymptotically stable in the mean-square sense if the solution of Eq. (2.10) satisfies

$$\lim_{n \rightarrow \infty} \sum_{\varpi=0}^{\ell} \mathbf{E} \|\mathbf{x}_n^{[\varpi]}\|^2 = 0.$$

Furthermore, Eq. (2.10) is said to be globally exponentially stable in the mean-square sense if the solution of Eq. (2.10) satisfies

$$\sum_{\varpi=0}^{\ell} \mathbf{E} \|\mathbf{x}_n^{[\varpi]}\|^2 \leq \eta(1 - \tau)^n, \quad \forall n \in \mathbb{Z}_0,$$

where η, τ are constants, $\tau \in (0, 1)$.

Based on Definition 2.1, we give the following quasi-stabilized definitions for GONs (2.1) via the stability of T-S fuzzy GONs (2.5).

Definition 2.2. GONs (2.1) and (2.2) are called **globally asymptotically quasi-stabilized in the mean-square sense (MSGAQS)** if T-S fuzzy GONs (2.5) and (2.6) are globally asymptotically stable in the mean-square sense under the controller (2.8). Namely,

$$\lim_{n \rightarrow \infty} \sum_{i=1}^N \sum_{\varpi=0}^{\ell} \mathbf{E} \left[\|\mathbf{u}_{i,n}^{[\varpi]}\|^2 + \|\mathbf{v}_{i,n}^{[\varpi]}\|^2 \right] = 0.$$

Definition 2.3. GONs (2.1) and (2.2) are called **globally exponentially quasi-stabilized in the mean-square sense (MSGEQS)** if T-S fuzzy GONs (2.5) and (2.6) are globally exponentially stable in the mean-square sense under the controller (2.8). Namely, there exist constants $\tau \in (0, 1)$ and $\eta > 0$ so that

$$\sum_{i=1}^N \sum_{\varpi=0}^{\ell} \mathbf{E} \left[\|\mathbf{u}_{i,n}^{[\varpi]}\|^2 + \|\mathbf{v}_{i,n}^{[\varpi]}\|^2 \right] \leq \eta(1 - \tau)^n, \quad \forall n \in \mathbb{Z}_0.$$

Remark 2.10. It is noteworthy that GONs (2.3) are derived by an equivalent transformation of GONs (2.1), and T-S fuzzy GONs (2.5) can be regarded as an approximate model of GONs (2.3). The present study proposes the original concepts of quasi-stabilization for GONs (2.1) via the stability (see Definition 2.1) of T-S fuzzy GONs (2.5) under the boundary controls.

Lemma 2.1 ([51, Theorem 25, discrete formula of integration by parts]). Let $\{\mathbf{u}^{[\varpi]} \in \mathbb{R}^m : v \in [0, \ell]_{\mathbb{Z}}\}$ be the real-valued sequence and W be a m -order matrix, then

$$\sum_{\varpi=1}^{\ell-1} \mathbf{u}^{[\varpi]T} W \Delta_i^2 \mathbf{u}^{[\varpi-1]} = \frac{1}{i} \mathbf{u}^{[\varpi]T} W \Delta_i \mathbf{u}^{[\varpi-1]} \Big|_1^{\ell} - \sum_{\varpi=1}^{\ell-1} \Delta_i \mathbf{u}^{[\varpi]T} W \Delta_i \mathbf{u}^{[\varpi]}.$$

Lemma 2.2 ([52, Corollary 2, discrete inequalities of Wirtinger's type]). The inequalities below are valid for n -dimensional real-valued sequence $\{\mathbf{u}^{[\varpi]} \in \mathbb{R}^m : v \in [0, \ell]_{\mathbb{Z}}\}$ and m -order positive definitive matrix W . If $\mathbf{u}^{[0]} = 0$, then

$$\begin{aligned} 4 \sin^2 \frac{\pi}{2(\ell+1)} \sum_{\varpi=0}^{\ell} \mathbf{u}^{[\varpi]T} W \mathbf{u}^{[\varpi]} &\leq \ell^2 \sum_{\varpi=0}^{\ell-1} (\Delta_i \mathbf{u}^{[\varpi]})^T W \Delta_i \mathbf{u}^{[\varpi]} \\ &\leq 4 \cos^2 \frac{\pi}{2\ell+1} \sum_{\varpi=0}^{\ell} \mathbf{u}^{[\varpi]T} W \mathbf{u}^{[\varpi]}. \end{aligned}$$

If $\mathbf{u}^{[0]} = 0$ may not be valid, then

$$\ell^2 \sum_{\varpi=0}^{\ell-2} (\Delta_i \mathbf{u}^{[\varpi]})^T W \Delta_i \mathbf{u}^{[\varpi]} \leq 4 \cos^2 \frac{\pi}{2\ell} \sum_{\varpi=0}^{\ell-1} \mathbf{u}^{[\varpi]T} W \mathbf{u}^{[\varpi]}.$$

Remark 2.11. In the process of handling the diffusive term in T-S fuzzy GONs (2.5), the discrete formula of integration by parts (Lemma 2.1) plays an important role. In order to reduce the order of the operator Δ_i^2 , the discrete inequalities of Wirtinger's type in Lemma 2.2 fulfill an indispensable function.

Lemma 2.3 ([53]). If $A, B \in \mathbb{R}^m$, $W > 0$ is a m -order matrix and $\vartheta > 0$ is a number, then

$$A^T W B + B^T W A \leq \vartheta A^T W A + \frac{1}{\vartheta} B^T W B.$$

3. Quasi-stabilization via boundary controls

For simplification, let $A_o = \frac{I - e^{-Ah}}{A}$, $\tilde{A}_o = \frac{I - e^{-\tilde{A}h}}{\tilde{A}}$, and λ_B denote the maximum eigenvalue of the matrix $B_r^T B_r$ for all $r = 1, 2, \dots, L$. According to assumption (g_2) and (2.9), it concludes

$$\begin{cases} \Delta_i \mathbf{u}_{i,n}^{[\ell-1]} = \frac{1}{l} \left[\theta_{i,\alpha_n} \rho_{i,n}^{\alpha_n} - \mathbf{u}_{i,n}^{[\ell-1]} - \xi_i^* \right], \\ \Delta_i \mathbf{v}_{i,n}^{[\ell-1]} = \frac{1}{l} \left[\vartheta_{i,\alpha_n} \varrho_{i,n}^{\alpha_n} - \mathbf{v}_{i,n}^{[\ell-1]} - \zeta_i^* \right], \end{cases} \quad (3.1)$$

where $n \in \mathbb{Z}_0$, $i = 1, 2, \dots, N$.

Design the feedback controls in (2.9) by

$$\begin{cases} \rho_{i,n}^{\alpha_n} = \frac{1}{\theta_{i,\alpha_n}} (\mathbf{u}_{i,n}^{[\ell-1]} + \xi_i^*) + \frac{1}{\theta_{i,\alpha_n}} \sum_{\varpi=1}^{\ell-1} \Xi_i \mathbf{u}_{i,n}^{[\varpi]}, \\ \varrho_{i,n}^{\alpha_n} = \frac{1}{\vartheta_{i,\alpha_n}} (\mathbf{v}_{i,n}^{[\ell-1]} + \zeta_i^*) + \frac{1}{\vartheta_{i,\alpha_n}} \sum_{\varpi=1}^{\ell-1} \Psi_i \mathbf{v}_{i,n}^{[\varpi]}, \end{cases} \quad (3.2)$$

where Ξ_i, Ψ_i stand for the gains of the controllers, $n \in \mathbb{Z}_0$, $i = 1, 2, \dots, N$.

Substituting (3.2) into (3.1), it leads to

$$\Delta_i \mathbf{u}_{i,n}^{[\ell-1]} = \frac{1}{l} \sum_{\varpi=1}^{\ell-1} \Xi_i \mathbf{u}_{i,n}^{[\varpi]}, \quad \Delta_i \mathbf{v}_{i,n}^{[\ell-1]} = \frac{1}{l} \sum_{\varpi=1}^{\ell-1} \Psi_i \mathbf{v}_{i,n}^{[\varpi]}, \quad (3.3)$$

where $n \in \mathbb{Z}_0$, $i = 1, 2, \dots, N$.

Remark 3.1. While the majority of models in the real world are continuous, it is imperative to obtain the numerical solutions of these models during the computational process, that is, the discrete forms. From GONs (2.1) and (2.2), we can clearly view and calculate the states of the networks as the time n increases from $n = 0$ to $n = T > 0$, which is more intuitional than the continuous models. From (2.2) and the induction of (3.3), the Dirichlet boundary conditions can be turned into the Neumann case, which is rarely achieved in continuous models.

Theorem 3.1. Let assumptions (g_1) , (g_2) hold, the boundary controls be defined by (2.9) and (3.2), and the following conditions be fulfilled.

(g_3) It exists m -order matrices $P_i > 0, Q_i > 0, H_i > 0, Z_i > 0$ and number $\mathcal{d}_i > 0$ such that $A_o P_i A_o \leq \mathcal{d}_i I$,

$$\begin{aligned} \frac{12}{l^2} \cos^2 \frac{\pi}{2\ell} \Theta A_o P_i A_o \Theta + \frac{l^2}{4 \sin^2 \frac{\pi}{4\ell-2}} H_i &\leq \text{sym} (e^{-Ah} P_i A_o \Theta), \\ \frac{12}{l^2} \cos^2 \frac{\pi}{2\ell} \tilde{\Theta} \tilde{A}_o Q_i \tilde{A}_o \tilde{\Theta} + \frac{l^2}{4 \sin^2 \frac{\pi}{4\ell-2}} Z_i &\leq \text{sym} (e^{-\tilde{A}h} Q_i \tilde{A}_o \tilde{\Theta}), \end{aligned}$$

where $i = 1, 2, \dots, N$.

(g_4) It exists m -order matrices $\mathcal{U}_i, \mathcal{V}_i$ such that

$$\mathfrak{C}_i = \begin{bmatrix} \mathfrak{C}_{i,11} & \frac{1}{l^2} \mathcal{U}_i^T & \mathfrak{C}_{i,13} & 0 \\ * & -H_i & 0 & 0 \\ * & * & \mathfrak{C}_{i,33} & \frac{1}{l^2} \mathcal{V}_i^T \\ * & * & * & -Z_i \end{bmatrix} < 0,$$

where $\mathfrak{C}_{i,11} = -P_i + (1 + 2\varepsilon)e^{-Ah} P_i e^{-Ah} + 3\tilde{B}\tilde{A}_o Q_i \tilde{A}_o \tilde{B} + (3 + \frac{1}{\varepsilon})c^2 N \sum_{j=1}^N s_{ji}^2 \Pi^T A_o P_j A_o \Pi + \frac{1}{l^2} \text{sym}(\mathcal{U}_i)$, $\mathfrak{C}_{i,13} = (e^{-\tilde{A}h} Q_i \tilde{A}_o \tilde{B})^T$, $\mathfrak{C}_{i,33} = -Q_i + (1 + \varepsilon)e^{-\tilde{A}h} Q_i e^{-\tilde{A}h} + (3 + \frac{1}{\varepsilon})\lambda_B \mathcal{d}_i I + (3 + \frac{1}{\varepsilon})\tilde{c}^2 N \sum_{j=1}^N \tilde{s}_{ji}^2 \Pi^T \tilde{A}_o Q_j \tilde{A}_o \Pi + \frac{1}{l^2} \text{sym}(\mathcal{V}_i)$, $\varepsilon > 0$, $i = 1, 2, \dots, N$. Besides, the gains of the boundary controls (3.2) are defined by

$$\Xi_i = \text{inv}(e^{-Ah} P_i A_o \Theta) \mathcal{U}_i, \quad \Psi_i = \text{inv}(e^{-\tilde{A}h} Q_i \tilde{A}_o \tilde{\Theta}) \mathcal{V}_i, \quad i = 1, 2, \dots, N. \quad (3.4)$$

Then GONs (2.1) and (2.2) are **MSGAQS** (see Definition 2.2), i.e., T-S fuzzy GONs (2.5) and (2.6) are globally asymptotically stable in the mean-square sense under the boundary controls. Namely,

$$\lim_{n \rightarrow \infty} \sum_{i=1}^N \sum_{\varpi=0}^{\ell} \mathbf{E} \left[\|\mathbf{u}_{i,n}^{[\varpi]}\|^2 + \|\mathbf{v}_{i,n}^{[\varpi]}\|^2 \right] = 0.$$

Proof. Define a Lyapunov functional, which is described by

$$U_n = U_{1,n} + U_{2,n},$$

where

$$U_{1,n} = \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\mathbf{u}_{i,n}^{[\varpi]})^T P_i \mathbf{u}_{i,n}^{[\varpi]} \quad U_{2,n} = \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\mathbf{v}_{i,n}^{[\varpi]})^T Q_i \mathbf{v}_{i,n}^{[\varpi]}, \quad \forall n \in \mathbb{Z}_0.$$

Apparently,

$$\Delta U_{1,n} = U_{1,n+1} - U_{1,n} = \underbrace{\sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\mathbf{u}_{i,n+1}^{[\varpi]})^T P_i \mathbf{u}_{i,n+1}^{[\varpi]}}_{\mathfrak{R}_{1,n}} - \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\mathbf{u}_{i,n}^{[\varpi]})^T P_i \mathbf{u}_{i,n}^{[\varpi]}, \quad (3.5)$$

where $n \in \mathbb{Z}_0$. Based on T-S fuzzy GONs (2.5), it follows

$$\begin{aligned} \mathfrak{R}_{1,n} &\leq \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\mathbf{u}_{i,n}^{[\varpi]})^T e^{-Ah} P_i e^{-Ah} \mathbf{u}_{i,n}^{[\varpi]} \\ &\quad + \underbrace{\sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\Delta_{c,i}^2 \mathbf{u}_{i,n}^{[\varpi]})^T \Theta A_o P_i A_o \Theta \Delta_{c,i}^2 \mathbf{u}_{i,n}^{[\varpi]}}_{\mathbf{F}_{1,n}} \\ &\quad + \underbrace{\sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} \left[\sum_{r=1}^L \hat{h}_r(z_n^{[\varpi]}) B_r \mathbf{v}_{i,n}^{[\varpi]} \right]^T A_o P_i A_o \left[\sum_{r=1}^L \hat{h}_r(z_n^{[\varpi]}) B_r \mathbf{v}_{i,n}^{[\varpi]} \right]}_{\mathbf{F}_{2,n}} \\ &\quad + \underbrace{c^2 \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} \left[\sum_{j=1}^N s_{ij} \Pi \mathbf{u}_{j,n}^{[\varpi]} \right]^T A_o P_i A_o \left[\sum_{j=1}^N s_{ij} \Pi \mathbf{u}_{j,n}^{[\varpi]} \right]}_{\mathbf{F}_{3,n}} \\ &\quad + \underbrace{\text{sym} \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\mathbf{u}_{i,n}^{[\varpi]})^T e^{-Ah} P_i A_o \Theta \Delta_{c,i}^2 \mathbf{u}_{i,n}^{[\varpi]}}_{\mathbf{F}_{4,n}} \\ &\quad + \epsilon \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\mathbf{u}_{i,n}^{[\varpi]})^T e^{-Ah} P_i e^{-Ah} \mathbf{u}_{i,n}^{[\varpi]} + \frac{1}{\epsilon} \mathbf{F}_{2,n} \\ &\quad + \underbrace{c \text{sym} \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\mathbf{u}_{i,n}^{[\varpi]})^T e^{-Ah} P_i A_o \left[\sum_{j=1}^N s_{ij} \Pi \mathbf{u}_{j,n}^{[\varpi]} \right]}_{\mathbf{F}_{5,n}} \\ &\quad + \underbrace{\text{sym} \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\Delta_{c,i}^2 \mathbf{u}_{i,n}^{[\varpi]})^T \Theta A_o P_i A_o \left[\sum_{r=1}^L \hat{h}_r(z_n^{[\varpi]}) B_r \mathbf{v}_{i,n}^{[\varpi]} \right]}_{\mathbf{F}_{6,n}} \\ &\quad + \underbrace{c \text{sym} \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\Delta_{c,i}^2 \mathbf{u}_{i,n}^{[\varpi]})^T \Theta A_o P_i A_o \left[\sum_{j=1}^N s_{ij} \Pi \mathbf{u}_{j,n}^{[\varpi]} \right]}_{\mathbf{F}_{7,n}} \\ &\quad + \underbrace{c \text{sym} \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} \left[\sum_{r=1}^L \hat{h}_r(z_n^{[\varpi]}) B_r \mathbf{v}_{i,n}^{[\varpi]} \right]^T A_o P_i A_o \left[\sum_{j=1}^N s_{ij} \Pi \mathbf{u}_{j,n}^{[\varpi]} \right]}_{\mathbf{F}_{8,n}}, \end{aligned} \quad (3.6)$$

likewise,

$$\Delta U_{2,n} = U_{2,n+1} - U_{2,n} = \underbrace{\sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\mathbf{v}_{i,n+1}^{[\varpi]})^T Q_i \mathbf{v}_{i,n+1}^{[\varpi]}}_{\mathfrak{R}_{2,n}} - \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\mathbf{v}_{i,n}^{[\varpi]})^T Q_i \mathbf{v}_{i,n}^{[\varpi]}, \quad (3.7)$$

as well as

$$\begin{aligned} \mathfrak{R}_{2,n} &= \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\mathbf{v}_{i,n}^{[\varpi]})^T e^{-\tilde{A}h} Q_i e^{-\tilde{A}h} \mathbf{v}_{i,n}^{[\varpi]} \\ &\quad + \underbrace{\sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\Delta_{c,i}^2 \mathbf{v}_{i,n}^{[\varpi]})^T \tilde{\Theta} \tilde{A}_o Q_i \tilde{A}_o \tilde{\Theta} \Delta_{c,i}^2 \mathbf{v}_{i,n}^{[\varpi]}}_{\mathbf{G}_{1,n}} \\ &\quad + \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\mathbf{u}_{i,n}^{[\varpi]})^T \tilde{B} \tilde{A}_o Q_i \tilde{A}_o \tilde{B} \mathbf{u}_{i,n}^{[\varpi]} \\ &\quad + \underbrace{\tilde{c}^2 \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} \left[\sum_{j=1}^N \tilde{s}_{ij} \Pi \mathbf{v}_{j,n}^{[\varpi]} \right]^T \tilde{A}_o Q_i \tilde{A}_o \left[\sum_{j=1}^N \tilde{s}_{ij} \Pi \mathbf{v}_{j,n}^{[\varpi]} \right]}_{\mathbf{G}_{2,n}} \\ &\quad + \underbrace{\text{sym} \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\mathbf{v}_{i,n}^{[\varpi]})^T e^{-\tilde{A}h} Q_i \tilde{A}_o \tilde{\Theta} \Delta_{c,i}^2 \mathbf{v}_{i,n}^{[\varpi]}}_{\mathbf{G}_{3,n}} \\ &\quad + \text{sym} \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\mathbf{v}_{i,n}^{[\varpi]})^T e^{-\tilde{A}h} Q_i \tilde{A}_o \tilde{B} \mathbf{u}_{i,n}^{[\varpi]} \\ &\quad + \underbrace{\tilde{c} \text{sym} \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\mathbf{v}_{i,n}^{[\varpi]})^T e^{-\tilde{A}h} Q_i \tilde{A}_o \left[\sum_{j=1}^N \tilde{s}_{ij} \Pi \mathbf{v}_{j,n}^{[\varpi]} \right]}_{\mathbf{G}_{4,n}} \\ &\quad + \underbrace{\text{sym} \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\Delta_{c,i}^2 \mathbf{v}_{i,n}^{[\varpi]})^T \tilde{\Theta} \tilde{A}_o Q_i \tilde{A}_o \tilde{B} \mathbf{u}_{i,n}^{[\varpi]}}_{\mathbf{G}_{5,n}} \\ &\quad + \underbrace{\tilde{c} \text{sym} \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\Delta_{c,i}^2 \mathbf{v}_{i,n}^{[\varpi]})^T \tilde{\Theta} \tilde{A}_o Q_i \tilde{A}_o \left[\sum_{j=1}^N \tilde{s}_{ij} \Pi \mathbf{v}_{j,n}^{[\varpi]} \right]}_{\mathbf{G}_{6,n}} \\ &\quad + \underbrace{\tilde{c} \text{sym} \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\mathbf{u}_{i,n}^{[\varpi]})^T \tilde{B} \tilde{A}_o Q_i \tilde{A}_o \left[\sum_{j=1}^N \tilde{s}_{ij} \Pi \mathbf{v}_{j,n}^{[\varpi]} \right]}_{\mathbf{G}_{7,n}}, \quad \forall n \in \mathbb{Z}_0. \end{aligned} \quad (3.8)$$

On the basis of Lemma 2.3, it gets

$$\begin{cases} \mathbf{F}_{5,n} \leq \epsilon \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\mathbf{u}_{i,n}^{[\varpi]})^T e^{-Ah} P_i e^{-Ah} \mathbf{u}_{i,n}^{[\varpi]} + \frac{1}{\epsilon} \mathbf{F}_{3,n}, \\ \mathbf{F}_{6,n} \leq \mathbf{F}_{1,n} + \mathbf{F}_{2,n}, \quad \mathbf{F}_{7,n} \leq \mathbf{F}_{1,n} + \mathbf{F}_{3,n}, \quad \mathbf{F}_{8,n} \leq \mathbf{F}_{2,n} + \mathbf{F}_{3,n}, \end{cases} \quad (3.9)$$

as well as

$$\left\{ \begin{array}{l} \mathbf{G}_{6,n} \leq \mathbf{G}_{1,n} + \mathbf{G}_{2,n}, \\ \mathbf{G}_{4,n} \leq \varepsilon \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\mathbf{v}_{i,n}^{[\varpi]})^T e^{-\tilde{A}h} Q_i e^{-\tilde{A}h} \mathbf{v}_{i,n}^{[\varpi]} + \frac{1}{\varepsilon} \mathbf{G}_{2,n}, \\ \mathbf{G}_{5,n} \leq \mathbf{G}_{1,n} + \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\mathbf{u}_{i,n}^{[\varpi]})^T \tilde{B} \tilde{A}_o Q_i \tilde{A}_o \tilde{B} \mathbf{u}_{i,n}^{[\varpi]}, \\ \mathbf{G}_{7,n} \leq \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\mathbf{u}_{i,n}^{[\varpi]})^T \tilde{B} \tilde{A}_o Q_i \tilde{A}_o \tilde{B} \mathbf{u}_{i,n}^{[\varpi]} + \mathbf{G}_{2,n}, \quad \forall n \in \mathbb{Z}_0. \end{array} \right. \quad (3.10)$$

According to Lemma 2.2, it computes

$$\begin{aligned} 3\mathbf{F}_{1,n} &= 3 \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\Delta_{c,i}^2 \mathbf{u}_{i,n}^{[\varpi]})^T \Theta A_o P_i A_o \Theta \Delta_{c,i}^2 \mathbf{u}_{i,n}^{[\varpi]} \\ &= 3 \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} \Delta_i [\Delta_i (\mathbf{u}_{i,n}^{[\varpi-1]})^T] \Theta A_o P_i A_o \Theta \Delta_i (\Delta_i \mathbf{u}_{i,n}^{[\varpi-1]}) \\ &= 3 \sum_{i=1}^N \sum_{\varpi=0}^{\ell-2} \Delta_i [\Delta_i (\mathbf{u}_{i,n}^{[\varpi]})^T] \Theta A_o P_i A_o \Theta \Delta_i (\Delta_i \mathbf{u}_{i,n}^{[\varpi]}) \\ &\leq \frac{12}{l^2} \cos^2 \frac{\pi}{2\ell} \sum_{i=1}^N \sum_{\varpi=0}^{\ell-1} (\Delta_i \mathbf{u}_{i,n}^{[\varpi]})^T \Theta A_o P_i A_o \Theta \Delta_i \mathbf{u}_{i,n}^{[\varpi]}, \end{aligned} \quad (3.11)$$

and similarly,

$$\begin{aligned} 3\mathbf{G}_{1,n} &= 3 \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\Delta_{c,i}^2 \mathbf{v}_{i,n}^{[\varpi]})^T \tilde{\Theta} \tilde{A}_o Q_i \tilde{A}_o \tilde{\Theta} \Delta_{c,i}^2 \mathbf{v}_{i,n}^{[\varpi]} \\ &= 3 \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} \Delta_i [\Delta_i (\mathbf{v}_{i,n}^{[\varpi-1]})^T] \tilde{\Theta} \tilde{A}_o Q_i \tilde{A}_o \tilde{\Theta} \Delta_i (\Delta_i \mathbf{v}_{i,n}^{[\varpi-1]}) \\ &= 3 \sum_{i=1}^N \sum_{\varpi=0}^{\ell-2} \Delta_i [\Delta_i (\mathbf{v}_{i,n}^{[\varpi]})^T] \tilde{\Theta} \tilde{A}_o Q_i \tilde{A}_o \tilde{\Theta} \Delta_i (\Delta_i \mathbf{v}_{i,n}^{[\varpi]}) \\ &\leq \frac{12}{l^2} \cos^2 \frac{\pi}{2\ell} \sum_{i=1}^N \sum_{\varpi=0}^{\ell-1} \Delta_i (\mathbf{v}_{i,n}^{[\varpi]})^T \tilde{\Theta} \tilde{A}_o Q_i \tilde{A}_o \tilde{\Theta} \Delta_i \mathbf{v}_{i,n}^{[\varpi]}, \quad \forall n \in \mathbb{Z}_0. \end{aligned} \quad (3.12)$$

By right of Lemma 2.1 and boundary (2.6), it gains

$$\begin{aligned} \mathbf{F}_{4,n} &= \text{sym} \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\mathbf{u}_{i,n}^{[\varpi]})^T e^{-Ah} P_i A_o \Theta \Delta_{c,i}^2 \mathbf{u}_{i,n}^{[\varpi]} \\ &= \text{sym} \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\mathbf{u}_{i,n}^{[\varpi]})^T e^{-Ah} P_i A_o \Theta \Delta_i^2 \mathbf{u}_{i,n}^{[\varpi-1]} \\ &= \frac{1}{l} \text{sym} \sum_{i=1}^N (\mathbf{u}_{i,n}^{[\varpi]})^T e^{-Ah} P_i A_o \Theta \Delta_i \mathbf{u}_{i,n}^{[\varpi-1]} \Big|_{\varpi=1}^{\varpi=\ell} \\ &\quad - \text{sym} \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\Delta_i \mathbf{u}_{i,n}^{[\varpi]})^T e^{-Ah} P_i A_o \Theta \Delta_i \mathbf{u}_{i,n}^{[\varpi]} \\ &= \frac{1}{l} \text{sym} \sum_{i=1}^N (\mathbf{u}_{i,n}^{[\ell]})^T e^{-Ah} P_i A_o \Theta \Delta_i \mathbf{u}_{i,n}^{[\ell-1]} \\ &\quad - \frac{1}{l} \text{sym} \sum_{i=1}^N (\mathbf{u}_{i,n}^{[1]})^T e^{-Ah} P_i A_o \Theta \Delta_i \mathbf{u}_{i,n}^{[0]} \\ &\quad - \text{sym} \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\Delta_i \mathbf{u}_{i,n}^{[\varpi]})^T e^{-Ah} P_i A_o \Theta \Delta_i \mathbf{u}_{i,n}^{[\varpi]} \end{aligned}$$

$$\begin{aligned}
&= \underbrace{\frac{1}{l} \text{sym} \sum_{i=1}^N (\mathbf{u}_{i,n}^{[\ell]})^T e^{-Ah} P_i A_o \Theta \Delta_i \mathbf{u}_{i,n}^{[\ell-1]}}_{\mathbf{F}_{9,n}} \\
&\quad - \text{sym} \sum_{i=1}^N \sum_{\varpi=0}^{\ell-1} (\Delta_i \mathbf{u}_{i,n}^{[\varpi]})^T e^{-Ah} P_i A_o \Theta \Delta_i \mathbf{u}_{i,n}^{[\varpi]}, \quad \forall n \in \mathbb{Z}_0.
\end{aligned} \tag{3.13}$$

In the same way, it gets

$$\begin{aligned}
\mathbf{G}_{3,n} &= \text{sym} \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\mathbf{v}_{i,n}^{[\varpi]})^T e^{-\tilde{A}h} Q_i \tilde{A}_o \tilde{\Theta} \Delta_i^2 \mathbf{v}_{i,n}^{[\varpi-1]} \\
&= \frac{1}{l} \text{sym} \sum_{i=1}^N (\mathbf{v}_{i,n}^{[\varpi]})^T e^{-\tilde{A}h} Q_i \tilde{A}_o \tilde{\Theta} \Delta_i \mathbf{v}_{i,n}^{[\varpi-1]} \Big|_{\varpi=1}^{\varpi=\ell} \\
&\quad - \text{sym} \sum_{i=1}^N (\Delta_i \mathbf{v}_{i,n}^{[\varpi]})^T e^{-\tilde{A}h} Q_i \tilde{A}_o \tilde{\Theta} \Delta_i \mathbf{v}_{i,n}^{[\varpi]} \\
&= \frac{1}{l} \text{sym} \sum_{i=1}^N (\mathbf{v}_{i,n}^{[\ell]})^T e^{-\tilde{A}h} Q_i \tilde{A}_o \tilde{\Theta} \Delta_i \mathbf{v}_{i,n}^{[\ell-1]} \\
&\quad - \frac{1}{l} \text{sym} \sum_{i=1}^N (\mathbf{v}_{i,n}^{[1]})^T e^{-\tilde{A}h} Q_i \tilde{A}_o \tilde{\Theta} \Delta_i \mathbf{v}_{i,n}^{[0]} \\
&\quad - \text{sym} \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\Delta_i \mathbf{v}_{i,n}^{[\varpi]})^T e^{-\tilde{A}h} Q_i \tilde{A}_o \tilde{\Theta} \Delta_i \mathbf{v}_{i,n}^{[\varpi]} \\
&= \underbrace{\frac{1}{l} \text{sym} \sum_{i=1}^N (\mathbf{v}_{i,n}^{[\ell]})^T e^{-\tilde{A}h} Q_i \tilde{A}_o \tilde{\Theta} \Delta_i \mathbf{v}_{i,n}^{[\ell-1]}}_{\mathbf{G}_{8,n}} \\
&\quad - \text{sym} \sum_{i=1}^N \sum_{\varpi=0}^{\ell-1} (\Delta_i \mathbf{v}_{i,n}^{[\varpi]})^T e^{-\tilde{A}h} Q_i \tilde{A}_o \tilde{\Theta} \Delta_i \mathbf{v}_{i,n}^{[\varpi]}.
\end{aligned} \tag{3.14}$$

Via the generalized C_2 inequality, we have

$$\begin{aligned}
(3 + \frac{1}{\varepsilon}) \mathbf{F}_{2,n} &= (3 + \frac{1}{\varepsilon}) \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} \left[\sum_{r=1}^L \tilde{h}_r(z_n^{[\varpi]}) B_r \mathbf{v}_{i,n}^{[\varpi]} \right]^T A_o P_i A_o \left[\sum_{r=1}^L \tilde{h}_r(z_n^{[\varpi]}) B_r \mathbf{v}_{i,n}^{[\varpi]} \right] \\
&\leq (3 + \frac{1}{\varepsilon}) \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} \sum_{r=1}^L \tilde{h}_r(z_n^{[\varpi]}) \mathcal{A}_i(\mathbf{v}_{i,n}^{[\varpi]})^T B_r^T B_r \mathbf{v}_{i,n}^{[\varpi]} \\
&\leq (3 + \frac{1}{\varepsilon}) \lambda_B \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} \mathcal{A}_i(\mathbf{v}_{i,n}^{[\varpi]})^T \mathbf{v}_{i,n}^{[\varpi]},
\end{aligned} \tag{3.15}$$

as well as

$$\begin{aligned}
(3 + \frac{1}{\varepsilon}) \mathbf{F}_{3,n} &= (3 + \frac{1}{\varepsilon}) c^2 \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} \left[\sum_{j=1}^N s_{ij} \Pi \mathbf{u}_{j,n}^{[\varpi]} \right]^T A_o P_i A_o \left[\sum_{j=1}^N s_{ij} \Pi \mathbf{u}_{j,n}^{[\varpi]} \right] \\
&\leq (3 + \frac{1}{\varepsilon}) c^2 N \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} \sum_{j=1}^N s_{ij}^2 (\mathbf{u}_{j,n}^{[\varpi]})^T \Pi^T A_o P_i A_o \Pi \mathbf{u}_{j,n}^{[\varpi]} \\
&= (3 + \frac{1}{\varepsilon}) c^2 N \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} \sum_{j=1}^N s_{ji}^2 (\mathbf{u}_{i,n}^{[\varpi]})^T \Pi^T A_o P_j A_o \Pi \mathbf{u}_{i,n}^{[\varpi]},
\end{aligned} \tag{3.16}$$

as well as

$$(3 + \frac{1}{\varepsilon}) \mathbf{G}_{2,n} = (3 + \frac{1}{\varepsilon}) \tilde{c}^2 \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} \left[\sum_{j=1}^N \tilde{s}_{ij} \Pi \mathbf{v}_{j,n}^{[\varpi]} \right]^T \tilde{A}_o Q_i \tilde{A}_o \left[\sum_{j=1}^N \tilde{s}_{ij} \Pi \mathbf{v}_{j,n}^{[\varpi]} \right]$$

$$\begin{aligned}
&\leq (3 + \frac{1}{\varepsilon}) \tilde{c}^2 N \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} \sum_{j=1}^N \tilde{s}_{ij}^2 (\mathbf{v}_{j,n}^{[\varpi]})^T \Pi^T \tilde{A}_o Q_i \tilde{A}_o \Pi \mathbf{v}_{j,n}^{[\varpi]} \\
&= (3 + \frac{1}{\varepsilon}) \tilde{c}^2 N \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} \sum_{j=1}^N \tilde{s}_{ji}^2 (\mathbf{v}_{i,n}^{[\varpi]})^T \Pi^T \tilde{A}_o Q_j \tilde{A}_o \Pi \mathbf{v}_{i,n}^{[\varpi]}, \quad \forall n \in \mathbb{Z}_0.
\end{aligned} \tag{3.17}$$

Define $\tilde{\mathbf{u}}_{i,n}^{[\varpi]} = \mathbf{u}_{i,n}^{[\ell]} - \mathbf{u}_{i,n}^{[\varpi]}$ and $\tilde{\mathbf{v}}_{i,n}^{[\varpi]} = \mathbf{v}_{i,n}^{[\ell]} - \mathbf{v}_{i,n}^{[\varpi]}$ for $(\varpi, n) \in [0, \ell]_{\mathbb{Z}} \times \mathbb{Z}_0$, $i = 1, 2, \dots, N$. Relying on (3.3), it gains

$$\begin{aligned}
\mathbf{E}(\mathbf{F}_{9,n}) &= \frac{1}{l} \text{sym} \sum_{i=1}^N (\mathbf{u}_{i,n}^{[\ell]})^T e^{-Ah} P_i A_o \Theta \Delta_i \mathbf{u}_{i,n}^{[\ell-1]} \\
&= \frac{1}{l^2} \text{sym} \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\mathbf{u}_{i,n}^{[\ell]})^T e^{-Ah} P_i A_o \Theta \Xi_i \mathbf{u}_{i,n}^{[\varpi]} \\
&= \frac{1}{l^2} \text{sym} \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\tilde{\mathbf{u}}_{i,n}^{[\varpi]})^T \mathcal{Z}_i \mathbf{u}_{i,n}^{[\varpi]} + \frac{1}{l^2} \text{sym} \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\mathbf{u}_{i,n}^{[\varpi]})^T \mathcal{Z}_i \mathbf{u}_{i,n}^{[\varpi]},
\end{aligned} \tag{3.18}$$

similarly,

$$\begin{aligned}
\mathbf{E}(\mathbf{G}_{8,n}) &= \frac{1}{l} \text{sym} \sum_{i=1}^N (\mathbf{v}_{i,n}^{[\ell]})^T e^{-\tilde{A}h} Q_i \tilde{A}_o \tilde{\Theta} \tilde{\Delta}_i \mathbf{v}_{i,n}^{[\ell-1]} \\
&= \frac{1}{l^2} \text{sym} \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\mathbf{v}_{i,n}^{[\ell]})^T e^{-\tilde{A}h} Q_i \tilde{A}_o \tilde{\Theta} \Psi_i \mathbf{v}_{i,n}^{[\varpi]} \\
&= \frac{1}{l^2} \text{sym} \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\tilde{\mathbf{v}}_{i,n}^{[\varpi]})^T \mathcal{Z}_i \mathbf{v}_{i,n}^{[\varpi]} + \frac{1}{l^2} \text{sym} \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\mathbf{v}_{i,n}^{[\varpi]})^T \mathcal{Z}_i \mathbf{v}_{i,n}^{[\varpi]}
\end{aligned} \tag{3.19}$$

for every $n \in \mathbb{Z}_0$. Furthermore,

$$\begin{aligned}
0 &= \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\tilde{\mathbf{u}}_{i,n}^{[\varpi]})^T H_i \tilde{\mathbf{u}}_{i,n}^{[\varpi]} - \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\tilde{\mathbf{u}}_{i,n}^{[\varpi]})^T H_i \tilde{\mathbf{u}}_{i,n}^{[\varpi]} \\
&\leq \sum_{i=1}^N \sum_{\varpi=1}^{\ell} (\tilde{\mathbf{u}}_{i,n}^{[\varpi]})^T H_i \tilde{\mathbf{u}}_{i,n}^{[\varpi]} - \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\tilde{\mathbf{u}}_{i,n}^{[\varpi]})^T H_i \tilde{\mathbf{u}}_{i,n}^{[\varpi]} \\
&\stackrel{\varpi=\ell-\varpi'}{=} \sum_{i=1}^N \sum_{\varpi'=0}^{\ell-1} (\tilde{\mathbf{u}}_{i,n}^{[\ell-\varpi']})^T H_i \tilde{\mathbf{u}}_{i,n}^{[\ell-\varpi']} - \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\tilde{\mathbf{u}}_{i,n}^{[\varpi]})^T H_i \tilde{\mathbf{u}}_{i,n}^{[\varpi]} \\
&\leq \frac{l^2}{4 \sin^2 \frac{\pi}{4\ell-2}} \sum_{i=1}^N \sum_{\varpi'=0}^{\ell-2} (\Delta_i \tilde{\mathbf{u}}_{i,n}^{[\ell-\varpi']})^T H_i \Delta_i \tilde{\mathbf{u}}_{i,n}^{[\ell-\varpi']} - \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\tilde{\mathbf{u}}_{i,n}^{[\varpi]})^T H_i \tilde{\mathbf{u}}_{i,n}^{[\varpi]} \\
&\stackrel{\varpi'=\ell-\varpi}{=} \frac{l^2}{4 \sin^2 \frac{\pi}{4\ell-2}} \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\Delta_i \tilde{\mathbf{u}}_{i,n}^{[\varpi]})^T H_i \Delta_i \tilde{\mathbf{u}}_{i,n}^{[\varpi]} - \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\tilde{\mathbf{u}}_{i,n}^{[\varpi]})^T H_i \tilde{\mathbf{u}}_{i,n}^{[\varpi]} \\
&= \frac{l^2}{4 \sin^2 \frac{\pi}{4\ell-2}} \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\Delta_i \mathbf{u}_{i,n}^{[\varpi]})^T H_i \Delta_i \mathbf{u}_{i,n}^{[\varpi]} - \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\tilde{\mathbf{u}}_{i,n}^{[\varpi]})^T H_i \tilde{\mathbf{u}}_{i,n}^{[\varpi]},
\end{aligned} \tag{3.20}$$

and in like manner,

$$\begin{aligned}
0 &= \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\tilde{\mathbf{v}}_{i,n}^{[\varpi]})^T Z_i \tilde{\mathbf{v}}_{i,n}^{[\varpi]} - \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\tilde{\mathbf{v}}_{i,n}^{[\varpi]})^T Z_i \tilde{\mathbf{v}}_{i,n}^{[\varpi]} \\
&\leq \frac{l^2}{4 \sin^2 \frac{\pi}{4\ell-2}} \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\Delta_i \mathbf{v}_{i,n}^{[\varpi]})^T Z_i \Delta_i \mathbf{v}_{i,n}^{[\varpi]} - \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\tilde{\mathbf{v}}_{i,n}^{[\varpi]})^T Z_i \tilde{\mathbf{v}}_{i,n}^{[\varpi]}, \quad \forall n \in \mathbb{Z}_0.
\end{aligned} \tag{3.21}$$

Define $\sigma_i = (\mathbf{u}_i^T, \tilde{\mathbf{u}}_i^T, \mathbf{v}_i^T, \tilde{\mathbf{v}}_i^T)^T$ for $i = 1, 2, \dots, N$. Therefore, via (3.5)-(3.21), it summarizes

$$\mathbf{E}(\Delta U_n) \leq \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\sigma_{i,n}^{[\varpi]})^T \mathfrak{E}_i \sigma_{i,n}^{[\varpi]} \leq \lambda_0 \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} \mathbf{E} \|\sigma_{i,n}^{[\varpi]}\|^2, \tag{3.22}$$

where λ_0 denotes the maximum eigenvalue of matrix $\mathfrak{E}_i$ for all $i = 1, 2, \dots, N$. Clearly, $\lambda_0 < 0$ according to assumption (g₄).

Summing both sides of (3.22) from $n = 0$ to $n = T - 1$, it results in

$$\mathbf{E}(U_T) - \mathbf{E}(U_0) \leq \lambda_0 \sum_{n=0}^{T-1} \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} \mathbf{E} \|\sigma_{i,n}^{[\varpi]}\|^2,$$

which is equal to

$$\sum_{n=0}^{T-1} \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} \mathbf{E} \|\sigma_{i,n}^{[\varpi]}\|^2 \leq -\frac{1}{\lambda_0} \mathbf{E}(U_0) < +\infty.$$

Let $T \rightarrow \infty$,

$$\sum_{n=0}^{\infty} \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} \mathbf{E} \|\sigma_{i,n}^{[\varpi]}\|^2 < +\infty.$$

As a result, $\lim_{n \rightarrow \infty} \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} \mathbf{E} \|\sigma_{i,n}^{[\varpi]}\|^2 = 0$. Together with the boundary values in (2.9), T-S fuzzy GONs (2.5) and (2.6) are globally asymptotically stable in the mean-square sense. That is, GONs (2.1) and (2.2) are MSGAQS. This completes the proof. $\square$

Theorem 3.2. Suppose that the boundary controls and the gains are defined by (2.9), (3.2) and (3.4) orderly. Let assumptions (g_1) , (g_2) , (g_3) and the following condition be fulfilled.

(g_5) It exists m -order matrices $\mathcal{U}_i, \mathcal{V}_i$ and $\tau \in (0, 1]$ such that $\mathbf{C}\mathbf{E}_i^* = \mathbf{C}\mathbf{E}_i + \tau \text{diag}(P_i, 0, Q_i, 0) < 0$, where $\mathbf{C}\mathbf{E}_i$ is defined as that in (g_4) , $i = 1, 2, \dots, N$.

Then GONs (2.1) and (2.2) are **MSGEQS** (see Definition 2.3), i.e., T-S fuzzy GONs (2.5) and (2.6) are globally exponentially stable in the mean-square sense under the boundary controls. Moreover, there exist constants $\tau \in (0, 1)$ and $\eta > 0$ so that

$$\sum_{i=1}^N \sum_{\varpi=0}^{\ell} \mathbf{E} \left[\|\mathbf{u}_{i,n}^{[\varpi]}\|^2 + \|\mathbf{v}_{i,n}^{[\varpi]}\|^2 \right] \leq \eta (1 - \tau)^n, \quad \forall n \in \mathbb{Z}_0.$$

Proof. By (3.22), it exists $\tau \in (0, 1]$ such that

$$\begin{aligned} \mathbf{E}(\Delta U_n) + \tau \mathbf{E}(U_n) &\leq \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\sigma_{i,n}^{[\varpi]})^T \mathbf{C}\mathbf{E}_i^* \sigma_{i,n}^{[\varpi]} \\ &\leq \lambda_* \mathbf{E} \sum_{i=1}^N \sum_{\varpi=1}^{\ell-1} (\sigma_{i,n}^{[\varpi]})^T \sigma_{i,n}^{[\varpi]}, \quad \forall n \in \mathbb{Z}_0, \end{aligned} \quad (3.23)$$

where λ_* denotes the maximum eigenvalue of matrix $\mathbf{C}\mathbf{E}_i^*$ for all $i = 1, 2, \dots, N$. Clearly, $\lambda_* < 0$ according to assumption (g_5) . Therefore,

$$\mathbf{E}(\Delta U_n) + \tau \mathbf{E}(U_n) \leq 0 \Rightarrow \mathbf{E}(\Delta U_n) \leq (1 - \tau)^n \mathbf{E}U_0,$$

which inhibits that

$$\sum_{i=1}^N \sum_{\varpi=0}^{\ell} \mathbf{E} \left[\|\mathbf{u}_{i,n}^{[\varpi]}\|^2 + \|\mathbf{v}_{i,n}^{[\varpi]}\|^2 \right] \leq \frac{\mathbf{E}U_0}{\lambda_{PQ}} (1 - \tau)^n, \quad \forall n \in \mathbb{Z}_0,$$

where λ_{PQ} denotes the minimum eigenvalue of P_i, Q_i for all $i = 1, 2, \dots, N$. Associating with the boundary values in (2.9), T-S fuzzy GONs (2.5) and (2.6) are globally exponentially mean-squared stable. That is, GONs (2.1) and (2.2) are MSGEQS. This completes the proof. $\square$

Remark 3.2. In comparison to the extant reports on GONs [10–12,17,27], the current manuscript presents certain advantages. Firstly, this paper considers discrete GONs in both the spatial and temporal fields, which represents a superior approach to the discrete-time GONs discussed in papers [10,17]. Secondly, the boundary controls are involved in the Dirichlet boundary conditions, which differ from the Neumann boundaries discussed in existing literature [11,27]. Finally, this article discusses the T-S fuzzy techniques in the GONs, which were not previously investigated in the context of space-time discrete GONs [11,12,27].

4. Numerical example

In recent decades, discrete-time GONs have been the focus of extensive research by numerous scholars. Table 1 compares the most relevant works of discrete-time GONs, with the aim of enabling readers to intuitively understand the effectiveness of the methods and results proposed in this paper.

Table 1

Comparisons of the most relevant studies of discrete-time GONs. Abbreviation: SEP-Stabilization of equilibrium point.

	Discrete space	T-S fuzzification	Switch	SEP
Refs. [8–10]	No	No	No	No
Ref. [17]	No	No	No	Yes
Refs. [11,12]	Yes	No	No	No
Current study	Yes	Yes	Yes	Yes

Table 2

The values of $\beta_{1,r}$ when $\mathcal{L} = 5$.

$\beta_{1,1}$	$\beta_{1,2}$	$\beta_{1,3}$	$\beta_{1,4}$	$\beta_{1,5}$	$\beta_{1,6}$	$\beta_{1,7}$	$\beta_{1,8}$	$\beta_{1,9}$	$\beta_{1,10}$	$\beta_{1,11}$
-0.041	-0.049	-0.059	-0.067	0	0.275	0.135	0.088	0.065	0.052	0.043

Table 3

The values of $\beta_{2,r}$ when $\mathcal{L} = 5$.

$\beta_{2,1}$	$\beta_{2,2}$	$\beta_{2,3}$	$\beta_{2,4}$	$\beta_{2,5}$	$\beta_{2,6}$	$\beta_{2,7}$	$\beta_{2,8}$	$\beta_{2,9}$	$\beta_{2,10}$	$\beta_{2,11}$
-0.043	-0.051	-0.062	-0.072	0	0.300	0.144	0.093	0.068	0.054	0.044

Taking some specific parameters, let us consider a numerical example of the space-time discretized GONs as noted below

$$\left\{ \begin{array}{l} \xi_{i,n+1}^{[\varpi]} = \begin{pmatrix} e^{-2h} & 0 \\ 0 & e^{-h} \end{pmatrix} \xi_{i,n}^{[\varpi]} + \begin{pmatrix} 0.5(1-e^{-2h}) & 0 \\ 0 & 1-e^{-h} \end{pmatrix} \left[0.1\Delta_{c,i}^2 \xi_{i,n}^{[\varpi]} \right. \\ \quad \left. + \begin{pmatrix} 0.1 & 0 \\ 0 & 0.2 \end{pmatrix} \begin{pmatrix} g_1(\xi_{i1,n}^{[\varpi]}) \\ g_2(\xi_{i2,n}^{[\varpi]}) \end{pmatrix} + 0.05\xi_{1,n}^{[\varpi]} + 0.1\xi_{2,n}^{[\varpi]} + \begin{pmatrix} 14.454 \\ 11.2968 \end{pmatrix} \right], \\ \zeta_{i,n+1}^{[\varpi]} = \begin{pmatrix} e^{-4h} & 0 \\ 0 & e^{-5h} \end{pmatrix} \zeta_{i,n}^{[\varpi]} + \begin{pmatrix} 0.25(1-e^{-4h}) & 0 \\ 0 & 0.2(1-e^{-5h}) \end{pmatrix} \\ \quad \times \left[0.2\Delta_{c,i}^2 \zeta_{i,n}^{[\varpi]} + \begin{pmatrix} 0.5 & 0 \\ 0 & 0.25 \end{pmatrix} \xi_{i,n}^{[\varpi]} + 0.03\zeta_{1,n}^{[\varpi]} + 0.05\zeta_{2,n}^{[\varpi]} \right], \end{array} \right. \quad (4.1)$$

where $(\varpi, n) \in (0, 3)_{\mathbb{Z}} \times \mathbb{Z}_0$, $\xi_i = (\xi_{i1}, \xi_{i2})^T$, $\zeta_i = (\zeta_{i1}, \zeta_{i2})^T$, g_i denotes the nonlinear Hill function with uncertainty that satisfies $g_i(1) = 0.5$ and should be locally linearized via the T-S fuzzy method, $i = 1, 2$. $h = 0.1$, $\iota = 1$. Clearly, GONs (4.1) admits a unique equilibrium $(7.84, 13.408, 7.84, 13.408, 1, 1, 1, 1)^T$.

Letting $\mathbf{u}_i = (u_{i1}, u_{i2})^T$, $\mathbf{v}_i = (v_{i1}, v_{i2})^T$, $u_{i1} = \xi_{i1} - 7.84$, $u_{i2} = \xi_{i2} - 13.408$, $v_{i1} = \zeta_{i1} - 1$ and $v_{i2} = \zeta_{i2} - 1$, $i = 1, 2$. Thus, the following Takagi-Sugeno fuzzy approach for the Hill functions $\tilde{g}_1(v_{i1}) = g_1(\zeta_{i1}) - g_1(1)$ and $\tilde{g}_2(v_{i2}) = g_2(\zeta_{i2}) - g_2(1)$ in the discourse domain $[-10, 10]$ can be derived as shown below.

- **Model Rule r for Node i :** IF z is $M_r(z)$, THEN $g_i^\dagger(z) = \beta_{i,r}z$, where $M_r(s)$ represents the fuzzy set denoted by

$$M_r(s) = \begin{cases} 1 - \frac{|s - r\epsilon_*|}{\epsilon_*}, & \text{if } s \in ((r-1)\epsilon_*, (r+1)\epsilon_*), \\ 0, & \text{elsewhere,} \end{cases}$$

$\epsilon_* = \frac{10}{\mathcal{L}}$, $\mathcal{L}$ is a predetermined positive integer, $r \in [-\mathcal{L}, \mathcal{L}]_{\mathbb{Z}}$,

$$\beta_{i,r} = \begin{cases} 0, & \text{if } r = 0; \\ \frac{\tilde{g}_i(r\epsilon_*)}{r\epsilon_*}, & \text{if } r \in [-\mathcal{L}, \mathcal{L}]_{\mathbb{Z}} \setminus \{0\}. \end{cases}$$

Table 2-Table 7 display the values of the fuzzy coefficients $\beta_{i,r}$ for the T-S fuzzy system $g_i^\dagger(\cdot)$ above when $\mathcal{L} = 5, 15, 25$ orderly, $i = 1, 2$. Specifically, they correspond to the 11, 31, and 51 membership functions (MFs, see Fig. 1-Fig. 3), respectively. In consideration of the right subfigures in Fig. 1-Fig. 3, it can be observed that the T-S fuzzy systems demonstrate an enhanced precision in approximating the real data as the number of MFs increases.

Based on the GONs (4.1) and taking 51 MFs as shown in Fig. 3, a locally linear network, which is defuzzified by the Takagi-Sugeno fuzzy rules, is represented by

Table 4The values of $\beta_{1,r}$ when $\mathcal{L} = 15$.

$\beta_{1,1}$	$\beta_{1,2}$	$\beta_{1,3}$	$\beta_{1,4}$	$\beta_{1,5}$	$\beta_{1,6}$	$\beta_{1,7}$	$\beta_{1,8}$	$\beta_{1,9}$	$\beta_{1,10}$	$\beta_{1,11}$
-0.041	-0.044	-0.046	-0.049	-0.052	-0.055	-0.059	-0.062	-0.065	-0.067	-0.065
$\beta_{1,12}$	$\beta_{1,13}$	$\beta_{1,14}$	$\beta_{1,15}$	$\beta_{1,16}$	$\beta_{1,17}$	$\beta_{1,18}$	$\beta_{1,19}$	$\beta_{1,20}$	$\beta_{1,21}$	$\beta_{1,22}$
-0.051	0	0.202	0.405	0.275	0.205	0.163	0.135	0.115	0.100	0.088
$\beta_{1,23}$	$\beta_{1,24}$	$\beta_{1,25}$	$\beta_{1,26}$	$\beta_{1,27}$	$\beta_{1,28}$	$\beta_{1,29}$	$\beta_{1,30}$	$\beta_{1,31}$		
0.079	0.071	0.065	0.060	0.056	0.052	0.048	0.045	0.043		

Table 5The values of $\beta_{2,r}$ when $\mathcal{L} = 15$.

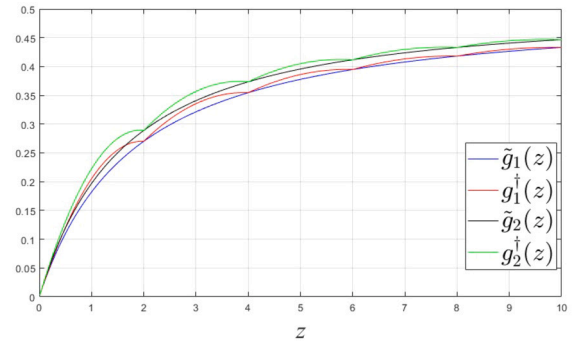
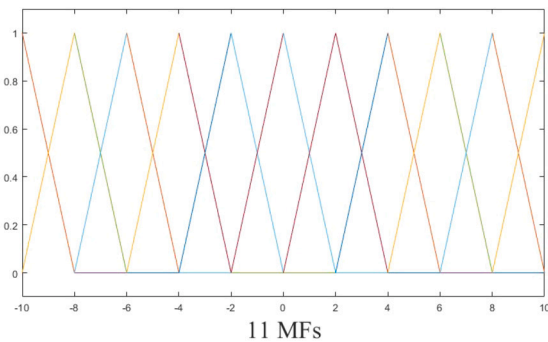
$\beta_{2,1}$	$\beta_{2,2}$	$\beta_{2,3}$	$\beta_{2,4}$	$\beta_{2,5}$	$\beta_{2,6}$	$\beta_{2,7}$	$\beta_{2,8}$	$\beta_{2,9}$	$\beta_{2,10}$	$\beta_{2,11}$
-0.043	-0.045	-0.048	-0.051	-0.054	-0.058	-0.062	-0.066	-0.069	-0.072	-0.070
$\beta_{2,12}$	$\beta_{2,13}$	$\beta_{2,14}$	$\beta_{2,15}$	$\beta_{2,16}$	$\beta_{2,17}$	$\beta_{2,18}$	$\beta_{2,19}$	$\beta_{2,20}$	$\beta_{2,21}$	$\beta_{2,22}$
-0.055	0	0.216	0.433	0.300	0.222	0.175	0.144	0.122	0.105	0.093
$\beta_{2,23}$	$\beta_{2,24}$	$\beta_{2,25}$	$\beta_{2,26}$	$\beta_{2,27}$	$\beta_{2,28}$	$\beta_{2,29}$	$\beta_{2,30}$	$\beta_{2,31}$		
0.083	0.075	0.068	0.063	0.058	0.054	0.050	0.047	0.044		

Table 6The values of $\beta_{1,r}$ when $\mathcal{L} = 25$.

$\beta_{1,1}$	$\beta_{1,2}$	$\beta_{1,3}$	$\beta_{1,4}$	$\beta_{1,5}$	$\beta_{1,6}$	$\beta_{1,7}$	$\beta_{1,8}$	$\beta_{1,9}$	$\beta_{1,10}$	$\beta_{1,11}$
-0.041	-0.043	-0.044	-0.046	-0.047	-0.049	-0.051	-0.052	-0.054	-0.056	-0.059
$\beta_{1,12}$	$\beta_{1,13}$	$\beta_{1,14}$	$\beta_{1,15}$	$\beta_{1,16}$	$\beta_{1,17}$	$\beta_{1,18}$	$\beta_{1,19}$	$\beta_{1,20}$	$\beta_{1,21}$	$\beta_{1,22}$
-0.061	-0.063	-0.065	-0.066	-0.067	-0.066	-0.063	-0.055	-0.038	0	0.085
$\beta_{1,23}$	$\beta_{1,24}$	$\beta_{1,25}$	$\beta_{1,26}$	$\beta_{1,27}$	$\beta_{1,28}$	$\beta_{1,29}$	$\beta_{1,30}$	$\beta_{1,31}$	$\beta_{1,32}$	$\beta_{1,33}$
0.295	0.443	0.342	0.275	0.228	0.195	0.170	0.150	0.135	0.122	0.111
$\beta_{1,34}$	$\beta_{1,35}$	$\beta_{1,36}$	$\beta_{1,37}$	$\beta_{1,38}$	$\beta_{1,39}$	$\beta_{1,40}$	$\beta_{1,41}$	$\beta_{1,42}$	$\beta_{1,43}$	$\beta_{1,44}$
0.102	0.095	0.088	0.082	0.077	0.073	0.069	0.065	0.062	0.059	0.056
$\beta_{1,45}$	$\beta_{1,46}$	$\beta_{1,47}$	$\beta_{1,48}$	$\beta_{1,49}$	$\beta_{1,50}$	$\beta_{1,51}$				
0.054	0.052	0.050	0.048	0.046	0.044	0.043				

Table 7The values of $\beta_{2,r}$ when $\mathcal{L} = 25$.

$\beta_{2,1}$	$\beta_{2,2}$	$\beta_{2,3}$	$\beta_{2,4}$	$\beta_{2,5}$	$\beta_{2,6}$	$\beta_{2,7}$	$\beta_{2,8}$	$\beta_{2,9}$	$\beta_{2,10}$	$\beta_{2,11}$
-0.043	-0.044	-0.046	-0.047	-0.049	-0.051	-0.053	-0.055	-0.057	-0.059	-0.062
$\beta_{2,12}$	$\beta_{2,13}$	$\beta_{2,14}$	$\beta_{2,15}$	$\beta_{2,16}$	$\beta_{2,17}$	$\beta_{2,18}$	$\beta_{2,19}$	$\beta_{2,20}$	$\beta_{2,21}$	$\beta_{2,22}$
-0.064	-0.066	-0.069	-0.071	-0.072	-0.071	-0.068	-0.060	-0.041	0	0.092
$\beta_{2,23}$	$\beta_{2,24}$	$\beta_{2,25}$	$\beta_{2,26}$	$\beta_{2,27}$	$\beta_{2,28}$	$\beta_{2,29}$	$\beta_{2,30}$	$\beta_{2,31}$	$\beta_{2,32}$	$\beta_{2,33}$
0.311	0.466	0.371	0.300	0.248	0.211	0.183	0.161	0.144	0.130	0.118
$\beta_{2,34}$	$\beta_{2,35}$	$\beta_{2,36}$	$\beta_{2,37}$	$\beta_{2,38}$	$\beta_{2,39}$	$\beta_{2,40}$	$\beta_{2,41}$	$\beta_{2,42}$	$\beta_{2,43}$	$\beta_{2,44}$
0.108	0.100	0.093	0.087	0.081	0.076	0.072	0.068	0.065	0.062	0.059
$\beta_{2,45}$	$\beta_{2,46}$	$\beta_{2,47}$	$\beta_{2,48}$	$\beta_{2,49}$	$\beta_{2,50}$	$\beta_{2,51}$				
0.056	0.054	0.051	0.049	0.048	0.046	0.044				

**Fig. 1.** 11 MFs and the T-S fuzzy systems of $\tilde{g}_1, \tilde{g}_2$.

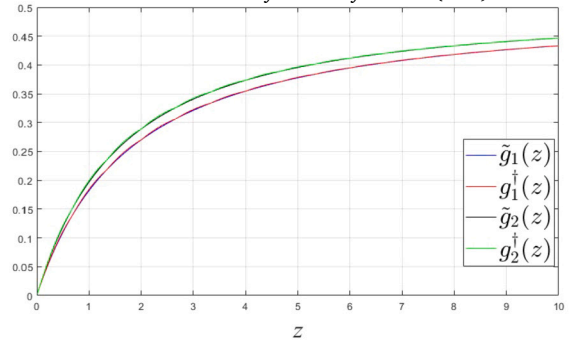
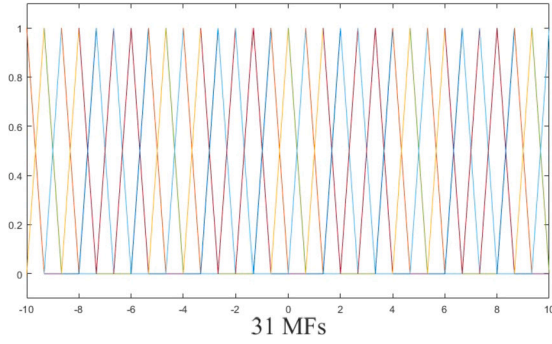


Fig. 2. 31 MFs and the T-S fuzzy systems of $\tilde{g}_1, \tilde{g}_2$.

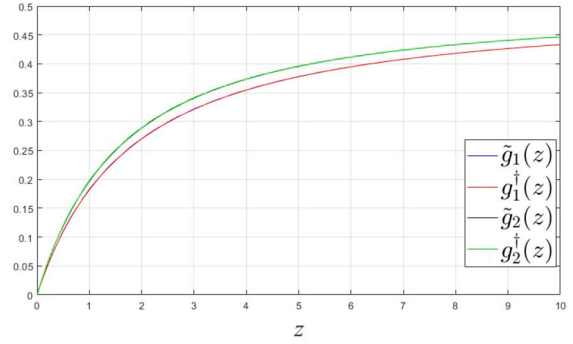
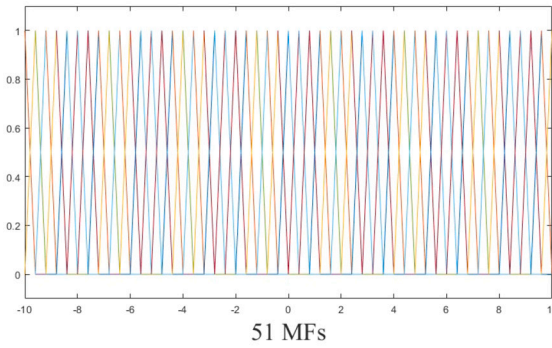


Fig. 3. 51 MFs and the T-S fuzzy systems of $\tilde{g}_1, \tilde{g}_2$.

$$\begin{cases} \mathbf{u}_{i,n+1}^{[\varpi]} = \begin{pmatrix} e^{-2h} & 0 \\ 0 & e^{-h} \end{pmatrix} \mathbf{u}_{i,n}^{[\varpi]} + \begin{pmatrix} 0.5(1-e^{-2h}) & 0 \\ 0 & 1-e^{-h} \end{pmatrix} \left[0.1\Delta_{c,i}^2 \mathbf{u}_{i,n}^{[\varpi]} \right. \\ \quad \left. + \sum_{r=-\mathcal{L}}^{\mathcal{L}} \tilde{h}_r^*(\zeta_i) \begin{pmatrix} 0.1\beta_{1,r}^* & 0 \\ 0 & 0.2\beta_{2,r}^* \end{pmatrix} \mathbf{v}_{i,n}^{[\varpi]} + 0.05\mathbf{u}_{1,n}^{[\varpi]} + 0.1\mathbf{u}_{2,n}^{[\varpi]} \right], \\ \mathbf{v}_{i,n+1}^{[\varpi]} = \begin{pmatrix} e^{-4h} & 0 \\ 0 & e^{-5h} \end{pmatrix} \mathbf{v}_{i,n}^{[\varpi]} + \begin{pmatrix} 0.25(1-e^{-4h}) & 0 \\ 0 & 0.2(1-e^{-5h}) \end{pmatrix} \\ \quad \times \left[0.2\Delta_{c,i}^2 \mathbf{v}_{i,n}^{[\varpi]} + \begin{pmatrix} 0.5 & 0 \\ 0 & 0.25 \end{pmatrix} \mathbf{u}_{i,n}^{[\varpi]} + 0.03\mathbf{v}_{1,n}^{[\varpi]} + 0.05\mathbf{v}_{2,n}^{[\varpi]} \right], \end{cases} \quad (4.2)$$

where $(\varpi, n) \in (0, 3)_{\mathbb{Z}} \times \mathbb{Z}_0$,

$$\tilde{h}_r^*(\zeta_i) = \frac{1}{3} \left[\frac{1}{51} + \tilde{h}_r(\zeta_{i1}) + \tilde{h}_r(\zeta_{i2}) \right], \quad \tilde{h}_r(\cdot) = M_r(\cdot) / \sum_{r=-25}^{25} M_r(\cdot),$$

$$\beta_{1,r}^* = \frac{\tilde{h}_r(\zeta_{i1})\beta_{1,r}}{\tilde{h}_r^*(\zeta_i)}, \quad \beta_{2,r}^* = \frac{\tilde{h}_r(\zeta_{i2})\beta_{2,r}}{\tilde{h}_r^*(\zeta_i)},$$

in which $i = 1, 2$, $r = -25, -24, \dots, 24, 25$. Taking the value of switching parameter α_n in the set $\{1, 2\}$ and

$$\mathbf{Pr}\{\alpha_n = 1\} = 0.3, \quad \mathbf{Pr}\{\alpha_n = 2\} = 0.7, \quad \forall n \in \mathbb{Z}_0.$$

Correspondingly, the implementation rates $\theta_{i,1} = 0.8$, $\theta_{i,2} = 0.9$, $\vartheta_{i,1} = 0.7$, $\vartheta_{i,2} = 0.85$, $i = 1, 2$. In accordance with the initial, boundary values in (2.6) and (2.9), the initial values and Dirichlet boundaries of T-S fuzzy GONs (4.2) are defined as

$$\begin{cases} \mathbf{u}_{i,0}^{[\varpi]} = (5i|\mathcal{r}_1| - 7.84, 5i|\mathcal{r}_2| - 13.408)^T, \\ \mathbf{v}_{i,0}^{[\varpi]} = (6i|\mathcal{r}_3| - 1, 6i|\mathcal{r}_4| - 1)^T, \quad \forall \varpi \in (0, 3)_{\mathbb{Z}}; \\ \mathbf{u}_{i,n}^{[0]} = 0, \quad \mathbf{v}_{i,n}^{[0]} = 0, \\ \mathbf{u}_{i,n}^{[3]} = \theta_{i,\alpha_n} \rho_{i,n}^{\alpha_n} - (7.84, 13.408)^T, \\ \mathbf{v}_{i,n}^{[3]} = \vartheta_{i,\alpha_n} \varrho_{i,n}^{\alpha_n} - (1, 1)^T, \quad \forall n \in \mathbb{Z}_0, i = 1, 2, \end{cases} \quad (4.3)$$

where $\mathcal{r}_1, \mathcal{r}_2, \mathcal{r}_3, \mathcal{r}_4$ represent four different normally distributed random numbers with the mean of 0 and variance of 1; $\rho_{i,n}^{\alpha_n}$ and $\varrho_{i,n}^{\alpha_n}$ are defined by

$$\begin{cases} \rho_{i,n}^{\alpha_n} = \frac{1}{\theta_{i,\alpha_n}} (\mathbf{u}_{i,n}^{[2]} + (7.84, 13.408)^T) + \frac{1}{\theta_{i,\alpha_n}} \sum_{\varpi=1}^2 \Xi_i \mathbf{u}_{i,n}^{[\varpi]}, \\ \varrho_{i,n}^{\alpha_n} = \frac{1}{\vartheta_{i,\alpha_n}} (\mathbf{v}_{i,n}^{[2]} + (1, 1)^T) + \frac{1}{\vartheta_{i,\alpha_n}} \sum_{\varpi=1}^2 \Psi_i \mathbf{v}_{i,n}^{[\varpi]}, \end{cases} \quad (4.4)$$

where Ξ_i, Ψ_i will be calculative below by some LMIs, $n \in \mathbb{Z}_0, i = 1, 2$.

Letting $\varepsilon = 0.1$. By computations, in Theorem 3.1, $\lambda_B = 0.078$, $\mathcal{d}_1 = 5.514$, $\mathcal{d}_2 = 5.482$,

$$\begin{aligned} P_1 &= \begin{pmatrix} 17.262 & 0 \\ 0 & 25.695 \end{pmatrix}, \quad P_2 = \begin{pmatrix} 17.311 & 0 \\ 0 & 26.689 \end{pmatrix}, \\ Q_1 &= \begin{pmatrix} 23.939 & 0 \\ 0 & 21.183 \end{pmatrix}, \quad Q_2 = \begin{pmatrix} 23.916 & 0 \\ 0 & 21.148 \end{pmatrix}, \\ H_1 &= \begin{pmatrix} 0.061 & 0 \\ 0 & 0.119 \end{pmatrix}, \quad H_2 = \begin{pmatrix} 0.061 & 0 \\ 0 & 0.125 \end{pmatrix}, \\ Z_1 &= \begin{pmatrix} 0.121 & 0 \\ 0 & 0.090 \end{pmatrix}, \quad Z_2 = \begin{pmatrix} 0.121 & 0 \\ 0 & 0.090 \end{pmatrix}, \\ \mathcal{U}_1 &= \begin{pmatrix} -0.418 & 0 \\ 0 & -0.229 \end{pmatrix}, \quad \mathcal{U}_2 = \begin{pmatrix} -0.416 & 0 \\ 0 & -0.226 \end{pmatrix}, \\ \mathcal{V}_1 &= \begin{pmatrix} -0.601 & 0 \\ 0 & -0.580 \end{pmatrix}, \quad \mathcal{V}_2 = \begin{pmatrix} -0.602 & 0 \\ 0 & -0.579 \end{pmatrix}. \end{aligned}$$

Via Eq. (3.4), it computes

$$\begin{aligned} \Xi_1 &= \begin{pmatrix} -3.266 & 0 \\ 0 & -1.036 \end{pmatrix}, \quad \Xi_2 = \begin{pmatrix} -3.235 & 0 \\ 0 & -0.982 \end{pmatrix}, \\ \Psi_1 &= \begin{pmatrix} -2.273 & 0 \\ 0 & -2.866 \end{pmatrix}, \quad \Psi_2 = \begin{pmatrix} -2.276 & 0 \\ 0 & -2.870 \end{pmatrix}. \end{aligned}$$

Through the strict verification, the assumptions in Theorem 3.1 hold. According to Theorem 3.1, the equilibrium point $(7.84, 13.408, 7.84, 13.408, 1, 1, 1, 1)^T$ of GONs (4.1) is globally asymptotically quasi-stabilized under the boundary controls as shown in (4.3)-(4.4).

Fig. 4-Fig. 11 picture the 3D and 2D global asymptotical stabilizations of the equilibrium point $(7.84, 13.408, 7.84, 13.408, 1, 1, 1, 1)^T$ of T-S fuzzy GONs (4.2) under the boundary controls as shown in (4.3)-(4.4) orderly. Fig. 12-Fig. 15 show the approximate trajectories of the states in the original GONs (4.1). Obviously, these approximate surfaces tend towards the equilibrium point $(7.84, 13.408, 7.84, 13.408, 1, 1, 1, 1)^T$ as the time n increases.

Remark 4.1. In papers [8,9,11,12], the authors engaged in discourse on the discrete-time GONs (see [8,9]) and space-time discrete GONs (see [11,12]). These works introduced some numerical examples with known Hill functions in the GONs. In contrast, the present study employs an unknown Hill function in the GONs via the T-S fuzzy technique, thereby extending and enhancing the contributions of papers [8,9,11,12].

5. Conclusions

This article considers the topic of global asymptotic or exponential stabilization of the equilibrium point of space-time discrete genetic oscillator networks with switching actuator failures and Dirichlet boundary controls. Via the technique of the Takagi-Sugeno fuzzification, the issues above can be transformed into the studies of global quasi-stabilization of the equilibrium point of the Takagi-Sugeno fuzzy genetic oscillator networks. With the helps of the method of the Lyapunov functional, the designation of boundary controls and the linear matrix inequalities, this article achieves the global asymptotic or exponential quasi-stabilization of the trivial

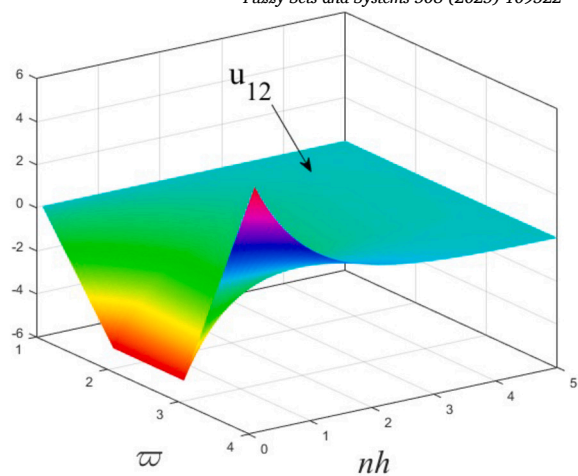
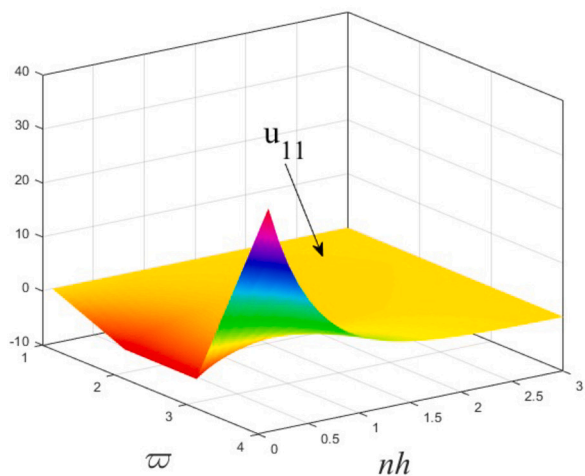


Fig. 4. MSGAQS of the states u_{11}, u_{12} in the T-S fuzzy GONs (4.2).

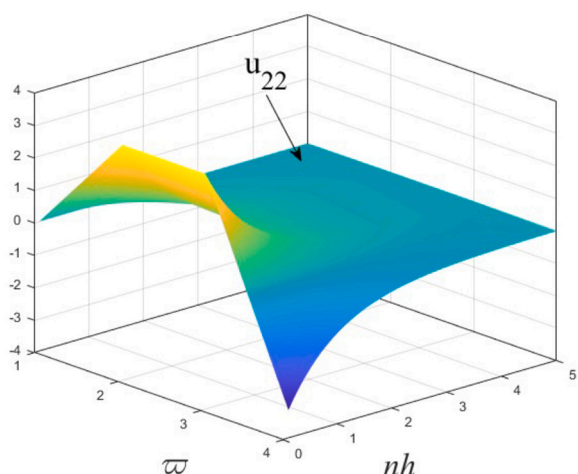
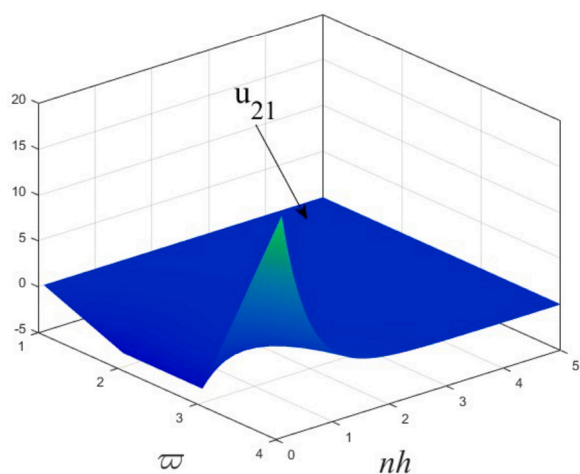


Fig. 5. MSGAQS of the states u_{21}, u_{22} in the T-S fuzzy GONs (4.2).

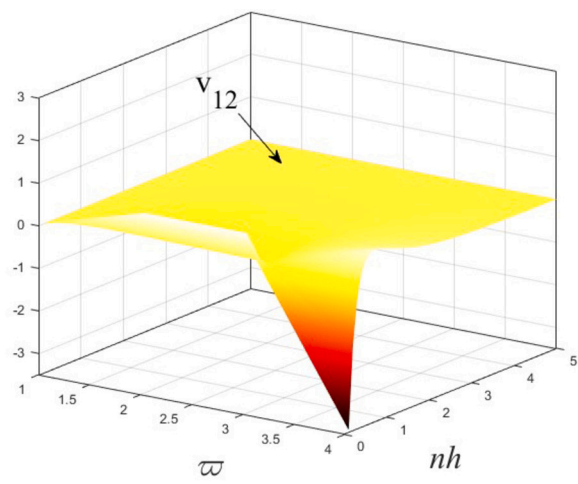
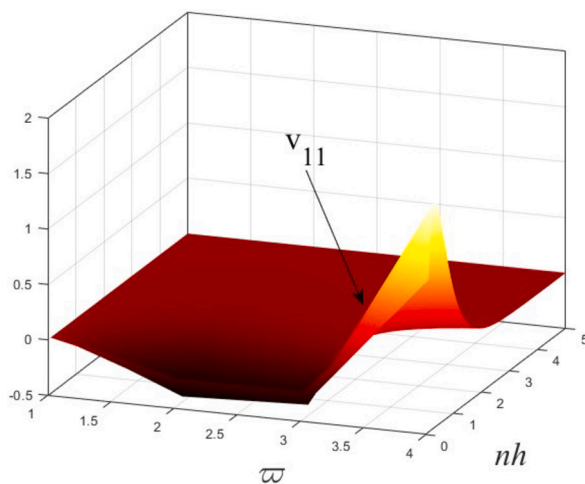


Fig. 6. MSGAQS of the states v_{11}, v_{12} in the T-S fuzzy GONs (4.2).

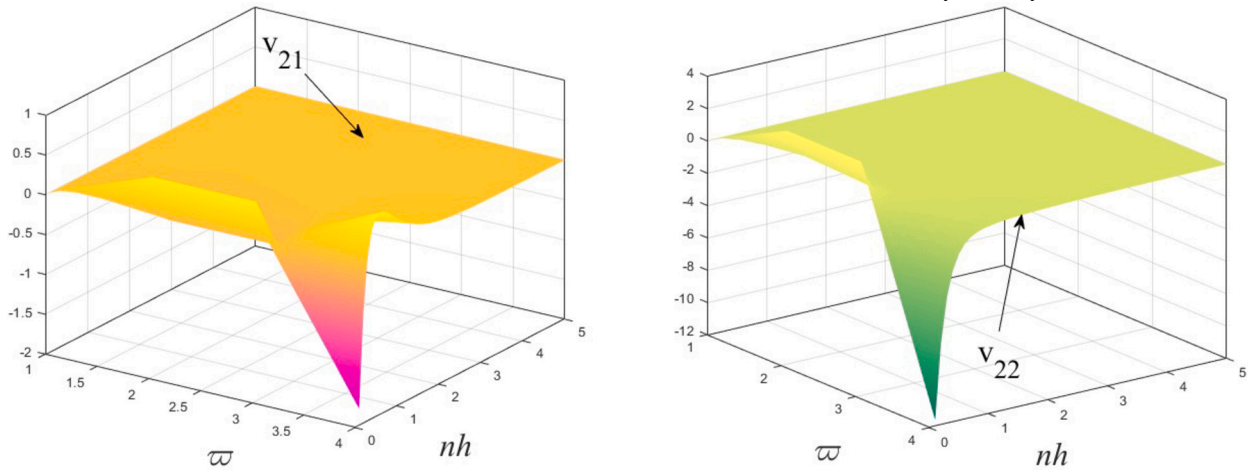


Fig. 7. MSGAQS of the states v_{21}, v_{22} in the T-S fuzzy GONs (4.2).

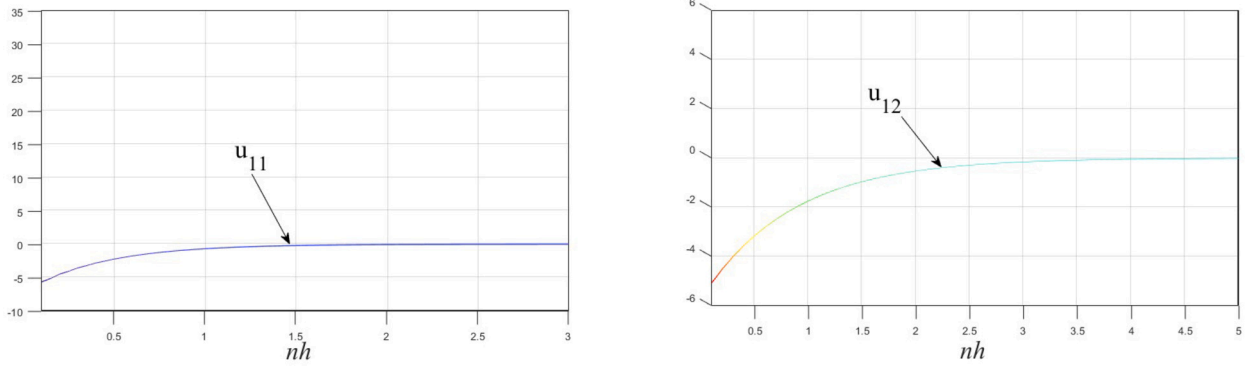


Fig. 8. The error curves u_{11}, u_{12} ($\varpi = 2$) of GONs (4.2) in 2D space.

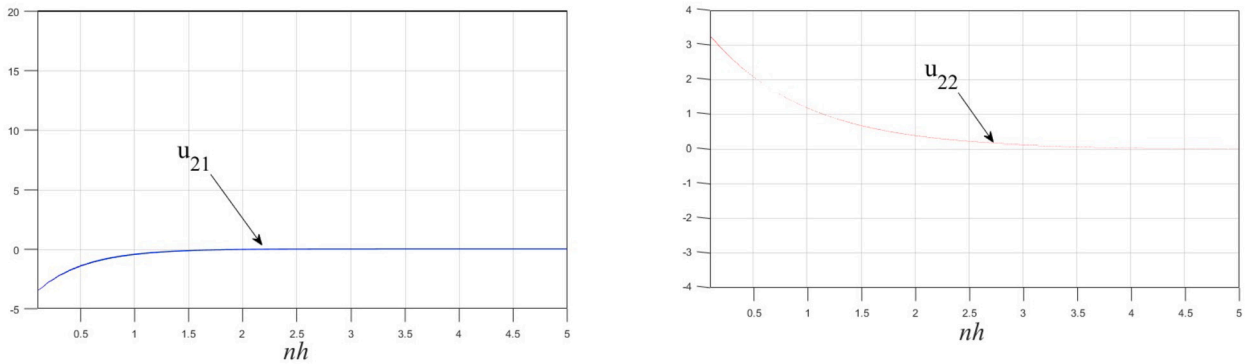


Fig. 9. The error curves u_{21}, u_{22} ($\varpi = 2$) of GONs (4.2) in 2D space.

equilibrium point for the space-time discrete genetic oscillator networks. The present study establishes a theoretical framework for the global stabilization of the equilibrium point to the genetic oscillator networks in the context of discrete schemes in space and time domains.

CRediT authorship contribution statement

Sufang Han: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Tianwei Zhang:** Writing – review & editing, Writing – original draft, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Yanning Wang:** Writing – review & editing, Validation.

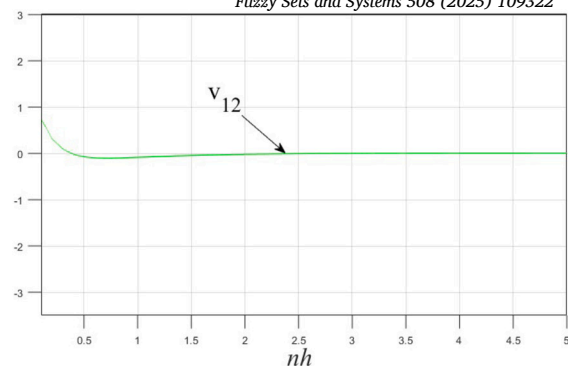
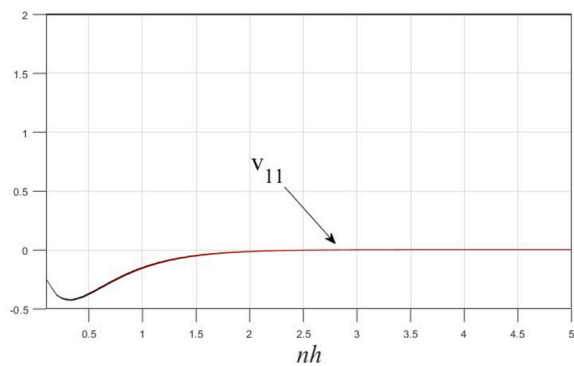


Fig. 10. The error curves v_{11}, v_{12} ($\varpi = 2$) of GONs (4.2) in 2D space.

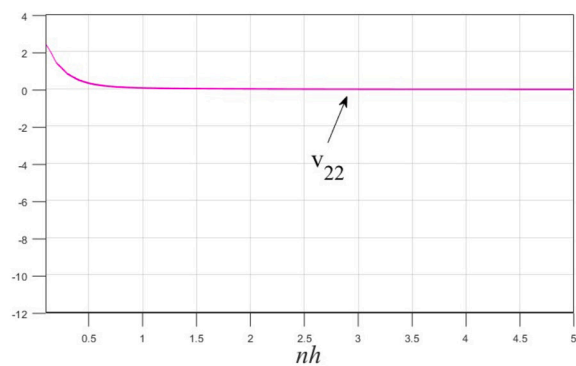
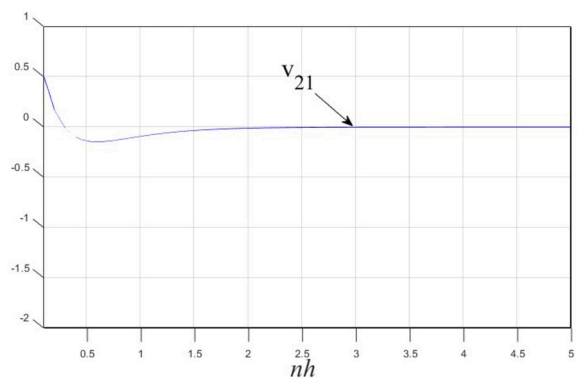


Fig. 11. The error curves v_{21}, v_{22} ($\varpi = 2$) of GONs (4.2) in 2D space.

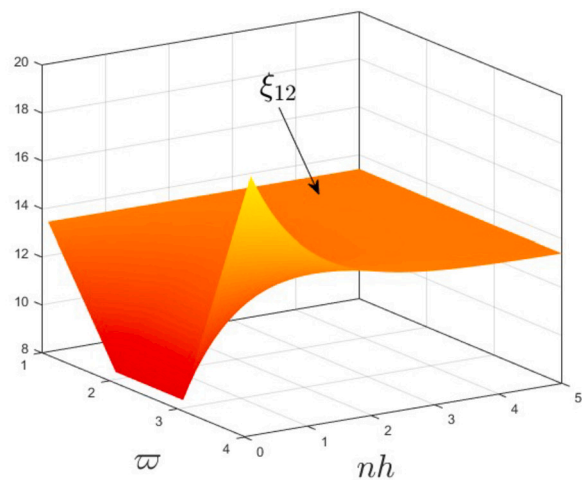
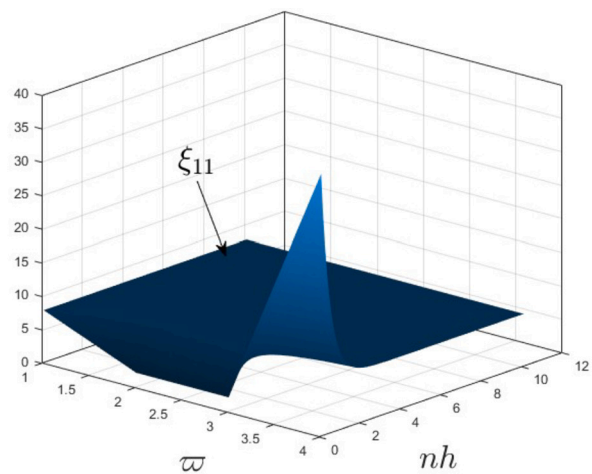


Fig. 12. The approximate surfaces of the states ξ_{11}, ξ_{12} in the GONs (4.1).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

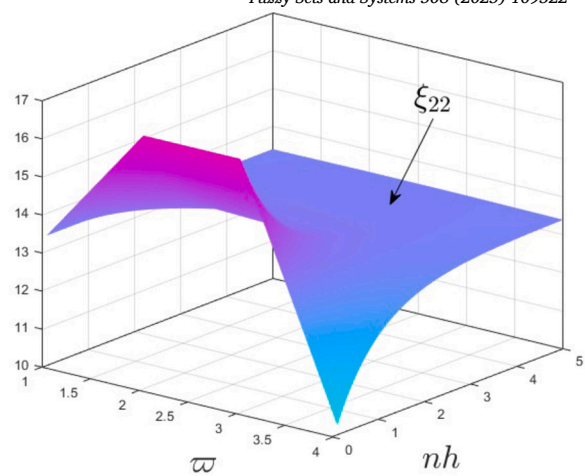
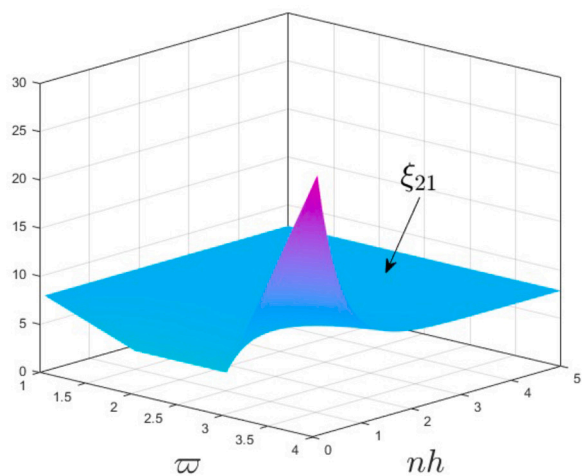


Fig. 13. The approximate surfaces of the states ξ_{21}, ξ_{22} in the GONs (4.1).

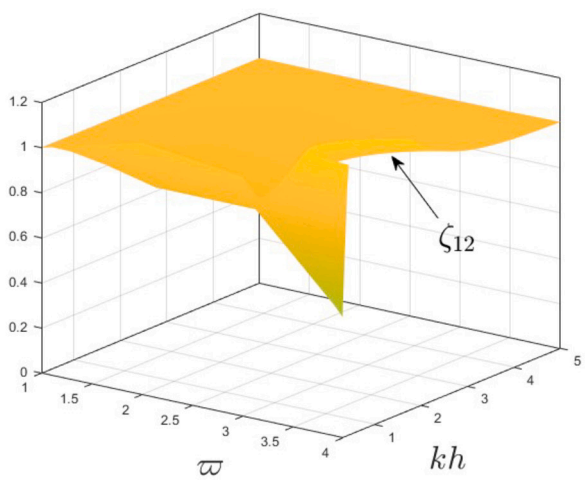
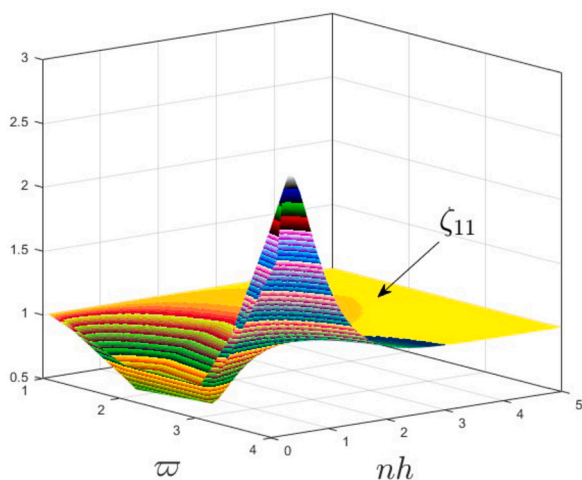


Fig. 14. The approximate surfaces of the states ζ_{11}, ζ_{12} in the GONs (4.1).

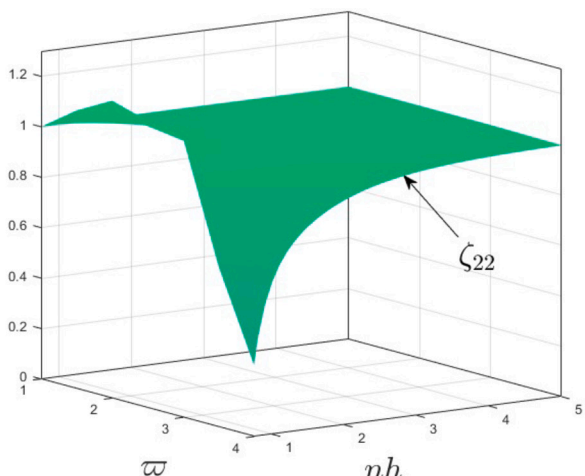
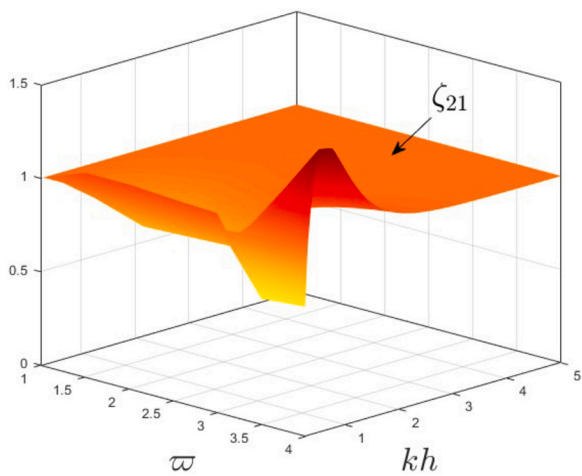


Fig. 15. The approximate surfaces of the states ζ_{21}, ζ_{22} in the GONs (4.1).

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Data availability

No data was used for the research described in the article.

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